Part Two

DATA SUFFICIENCY

Format of Data Sufficiency Questions

Most people have much more difficulty with the Data Sufficiency problems than with the Standard Math problems. However, the mathematical knowledge and skill required to solve Data Sufficiency problems is no greater than that required to solve standard math problems. What makes Data Sufficiency problems appear harder at first is the complicated directions. But once you become familiar with the directions, you'll find these problems no harder than standard math problems. In fact, people usually become proficient more quickly on Data Sufficiency problems.

The Directions

The directions for Data Sufficiency questions are rather complicated. Before reading any further, take some time to learn the directions cold. Some of the wording in the directions below has been changed from the GMAT to make it clearer. You should never have to look at the instructions during the test.

Directions: Each of the following Data Sufficiency problems contains a question followed by two statements, numbered (1) and (2). You need not solve the problem; rather you must decide whether the information given is sufficient to solve the problem.

The correct answer to a question is

A if statement (1) ALONE is sufficient to answer the question but statement (2) alone is not sufficient;
B if statement (2) ALONE is sufficient to answer the question but statement (1) alone is not sufficient;
C if the two statements TAKEN TOGETHER are sufficient to answer the question, but NEITHER statement ALONE is sufficient;
D if EACH statement ALONE is sufficient to answer the question;
E if the two statements TAKEN TOGETHER are still NOT sufficient to answer the question.

Numbers: Only real numbers are used. That is, there are no complex numbers.

Drawings: The drawings are drawn to scale according to the information given in the question, but may conflict with the information given in statements (1) and (2).

You can assume that a line that appears straight is straight and that angle measures cannot be zero.

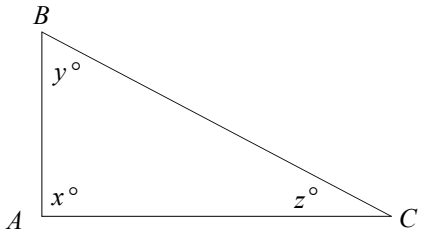
You can assume that the relative positions of points, angles, and objects are as shown.

All drawings lie in a plane unless stated otherwise.

Example:

In ΔABC to the right, what is the value of y ?

(1) $AB = AC$
(2) $x = 30$



Explanation: By statement (1), ΔABC is isosceles. Hence, its base angles are equal: $y = z$. Since the angle sum of a triangle is 180°, we get $x + y + z = 180$. Replacing z with y in this equation and then simplifying yields $x + 2y = 180$. Since statement (1) does not give a value for x, we cannot determine the value of y from statement (1) alone. By statement (2), $x = 30$. Hence, $x + y + z = 180$ becomes $30 + y + z = 180$, or $y + z = 150$. Since statement (2) does not give a value for z, we cannot determine the value of y from statement (2) alone. However, using both statements in combination, we can find both x and z and therefore y. Hence, the answer is C.

Notice in the above example that the triangle appears to be a right triangle. However, that cannot be assumed: angle A may be 89° or 91°, we can't tell from the drawing. **You must be very careful not to assume any more than what is explicitly given in a Data Sufficiency problem.**

Elimination

Data Sufficiency questions provide fertile ground for elimination. In fact, it is rare that you won't be able to eliminate some answer-choices. Remember, if you can eliminate at least one answer choice, the odds of gaining points by guessing are in your favor.

The following table summarizes how elimination functions with Data Sufficiency problems.

Statement	Choices Eliminated
(1) is sufficient	B, C, E
(1) is not sufficient	A, D
(2) is sufficient	A, C, E
(2) is not sufficient	B, D
(1) is not sufficient and (2) is not sufficient	A, B, D

Example 1: What is the 1st term in sequence S?

(1) The 3rd term of S is 4.
(2) The 2nd term of S is three times the 1st, and the 3rd term is four times the 2nd.

(1) is no help in finding the first term of S. For example, the following sequences each have 4 as their third term, yet they have different first terms:

0, 2, 4
–4, 0, 4

This eliminates choices A and D. Now, even if we are unable to solve this problem, we have significantly increased our chances of guessing correctly—from 1 in 5 to 1 in 3.

Turning to (2), we completely ignore the information in (1). Although (2) contains a lot of information, it also is not sufficient. For example, the following sequences each satisfy (2), yet they have different first terms:

1, 3, 12
3, 9, 36

This eliminates B, and our chances of guessing correctly have increased to 1 in 2.

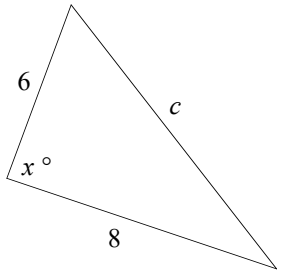
Next, we consider (1) and (2) together. From (1), we know *"the 3rd term of S is 4."* From (2), we know *"the 3rd term is four times the 2nd."* This is equivalent to saying the 2nd term is 1/4 the 3rd term: $\frac{1}{4}\cdot 4 \square 1$. Further, from (2), we know *"the 2nd term is three times the 1st."* This is equivalent to saying the 1st term is 1/3 the 2nd term: $\frac{1}{3}\cdot 1 \square \frac{1}{3}$. Hence, the first term of the sequence is fully determined:

$$\mathbf{\frac{1}{3}, 1, 4}$$

The answer is C.

Example 2: In the figure to the right, what is the area of the triangle?

(1) $c^2 = 6^2 + 8^2$
(2) $x = 90$



Recall that a triangle is a right triangle if and only if the square of the longest side is equal to the sum of the squares of the shorter sides (Pythagorean Theorem). Hence, (1) implies that the triangle is a right triangle. So the area of the triangle is $\frac{1}{2} \cdot 6 \cdot 8$. Note, there is no need to calculate the area—we just need to know that the area *can be* calculated. Hence, the answer is either A or D.

Turning to (2), we see immediately that we have a right triangle. Hence, again the area can be calculated. The answer is D.

Example 3: What is the value of $x - y$?

(1) $x + y = 3y - x$
(2) $x + y = x^3 + y^3$

Start with (1): $x + y = 3y - x$

Subtract $3y$ and add x to both sides of the equation: $2x - 2y = 0$

Divide by 2: $x - y = 0$

Hence, (1) is sufficient to determine the value of $x - y$, and therefore the answer is either A or D.

Turning to (2), we suspect there is not enough information since there are no like terms that can be combined as in (1). So use substitution to look for a counterexample. Let $x = y = 0$. Then $0 + 0 = 0^3 + 0^3$ and $x - y = 0$. However, if $x = 1$ and $y = 0$, then $1 + 0 = 1^3 + 0^3$ and $x - y = 1$. This shows that there are different pairs of numbers which satisfy (2) yet yield different values for $x - y$. Hence, (2) is not sufficient to determine the value of $x - y$. The answer is A. (Note, for the information to be sufficient, it is not enough to find *a* value of $x - y$; there must be a *unique* value.)

Example 4: If x is both the cube of an integer and between 2 and 200, what is the value of x?

(1) x is odd.
(2) x is the square of an integer.

Since x is both a cube and between 2 and 200, we are looking at the integers:

$$2^3, 3^3, 4^3, 5^3$$

which reduce to

$$8, 27, 64, 125$$

Since there are two odd integers in this set, (1) is not sufficient to uniquely determine the value of x. This eliminates choices A and D.

Next, there is only one perfect square, $64 = 8^2$, in the set. Hence, (2) is sufficient to deter mine the value of x. The answer is B.

Example 5: Is CAB a code word in language Q?

(1) ABC is the base word.
(2) If C immediately follows B, then C can be moved to the front of the code word to generate another word.

From (1), we cannot determine whether CAB is a code word since (1) gives no rule for generating another word from the base word. This eliminates A and D.

Turning to (2), we still cannot determine whether CAB is a code word since now we have no word to apply this rule to. This eliminates B.

However, if we consider (1) and (2) together, then we can determine whether CAB is a code word:

From (1), ABC is a code word.
From (2), the C in the code word ABC can be moved to the front of the word: CAB.

Hence, CAB is a code word and the answer is C.

Example 6: If x and y are positive integers, is $\sqrt{2xy}$ an integer?

(1) $y - x = 0$
(2) $xy = 1$ for some positive integer y.

From $y - x = 0$, we get $y = x$. Plug this into the expression $\sqrt{2xy}$:

$$\sqrt{2xx} \Rightarrow \sqrt{2x^2} \Rightarrow \sqrt{2}x$$

Since $\sqrt{2}$ is not an integer, $\sqrt{2xy}$ is not an integer. Hence, (1) is sufficient to answer the question.

Next, dividing both sides of $xy = 1$ by x yields $y = 1/x$. Plugging this into the expression $\sqrt{2xy}$ yields

$$\sqrt{2x\left(\frac{1}{x}\right)} \Rightarrow \sqrt{2}$$

Since $\sqrt{2}$ is not an integer, $\sqrt{2xy}$ is not an integer. Hence, (2) is also sufficient to answer the question. The answer is D.

Example 7: What is the value of $\frac{3+2p}{3p}$?

(1) $p^2 - 5p + 4 = 0$.
(2) The value of $\frac{1}{p} + \frac{2}{3}$ is –2.

Factoring (1) gives $(p - 1)(p - 4) = 0$.

Hence, $p - 1 = 0$ or $p - 4 = 0$.

So, $p = 1$ or $p = 4$.

However, if $p = 1$, then $\frac{3+2p}{3p} = \frac{5}{3}$. But if $p = 4$, then $\frac{3+2p}{3p} = \frac{11}{12}$. Therefore, (1) is not sufficient.

Translating (2) into an equation gives $\frac{1}{p} + \frac{2}{3} = -2$.

Adding the fractions on the left side gives $\frac{3+2p}{3p} = -2$. Hence, (2) is sufficient to answer the question.

The answer is B.

Example 8: If $x \neq -1$ and $y \neq \pm 1$, then what is the value of $\frac{xy - x}{(x + 1)(y - 1)}$?

(1) $x = 5$
(2) $y = 4$

Begin by factoring the expression: $\frac{x(y - 1)}{(x + 1)(y - 1)}$

Next, cancel the term $(y - 1)$: $\frac{x}{x + 1}$

Since x is the only variable left in the expression and (1) gives a value for x, we can calculate the value of the expression. Hence, (1) is sufficient to determine the value of the expression, and the answer is A or D.

Next, (2) is insufficient to determine the value of the expression since the value of x is not given. The answer is A.

Note, the value of y is not needed to determine the value of the expression since all the y's can be canceled from the expression; however, the value of x is needed since not all the x's can be canceled.

Unwarranted Assumptions

With data sufficiency problems, be careful not to read any more into a statement than what is given.

- **The main purpose of some difficult problems is to lure you into making an unwarranted assumption.**

If you avoid the temptation, these problems can become routine.

Example 6: Did Incumbent *I* get over 50% of the vote?

(1) Challenger *C* got 49% of the vote.
(2) Incumbent *I* got 25,000 of the 100,000 votes cast.

If you did not make any unwarranted assumptions, you probably did not find this to be a hard problem. What makes a problem difficult is not necessarily its underlying complexity; rather a problem is classified as difficult if many people miss it. A problem may be simple yet contain a psychological trap that causes people to answer it incorrectly.

The above problem is difficult because many people subconsciously assume that there are only two candidates. They then figure that since the challenger received 49% of the vote the incumbent received 51% of the vote. This would be a valid deduction if *C* were the only challenger*. But we cannot assume that. There may be two or more challengers. Hence, (1) is insufficient.

Now, consider (2) alone. Since Incumbent *I* received 25,000 of the 100,000 votes cast, *I* necessarily received 25% of the vote. Hence, the answer to the question is "No, the incumbent did not receive over 50% of the vote." Therefore, (2) is sufficient to answer the question. The answer is B.

Note, some people have trouble with (2) because they feel that the question asks for a "yes" answer. But on Data Sufficiency questions, a "no" answer is just as valid as a "yes" answer. What we're looking for is a *definite* answer.

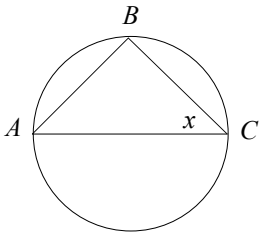
* You might ask, "What if some people voted for none-of-the-above?" But don't get carried away with finding exceptions. The writers of the GMAT would not set a trap that subtle.

Checking Extreme Cases



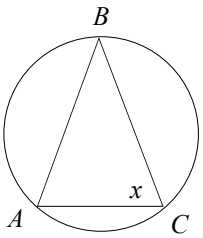
When Drawing a Geometric Figure or Checking a Given One, Be Sure to Include Drawings of Extreme Cases As Well As Ordinary Ones.

Example 1: In the figure to the right, AC is a chord and B is a point on the circle. What is the measure of angle x?

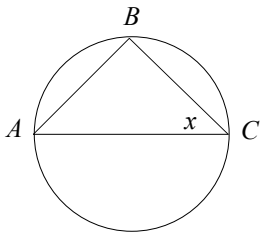


Although in the drawing AC looks to be a diameter, that cannot be assumed. All we know is that AC is a chord. Hence, numerous cases are possible, three of which are illustrated below:

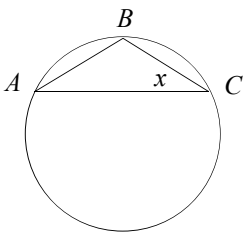
Case I



Case II



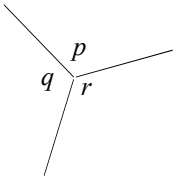
Case III



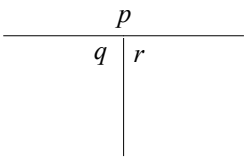
In Case I, x is greater than 45 degrees; in Case II, x equals 45 degrees; in Case III, x is less than 45 degrees. Hence, the given information is not sufficient to answer the question.

Example 2: Three rays emanate from a common point and form three angles with measures *p, q, and r*. What is the measure of $q + r$?

It is natural to make the drawing symmetric as follows:



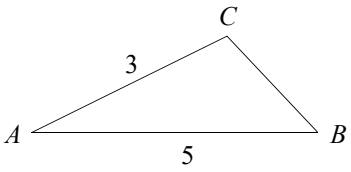
In this case, $p = q = r = 120^\circ$, so $q + r = 240^\circ$. However, there are other drawings possible. For example:



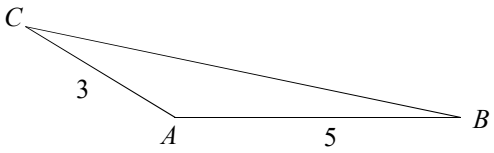
In this case, $q + r = 180^\circ$. Hence, the given information is not sufficient to answer the question.

Example 3: In triangle ABC, $AB = 5$ and $AC = 3$. Is $BC < 7$?

The most natural drawing is the following:

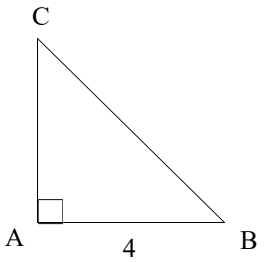


In this case, BC is less than 7. However, there is another drawing possible, as follows:



In this case, BC is greater than 7. Hence, the given information is not sufficient to answer the question.

Example 4: In the figure to the right, what is the area of triangle ΔABC ?



Although the drawing looks to be an isosceles triangle, that cannot be assumed. We are not given the length of side AC: it could be 4 units long or 100 units long, we don't know. Hence, the given information is not sufficient to answer the question.

Problem Set Z

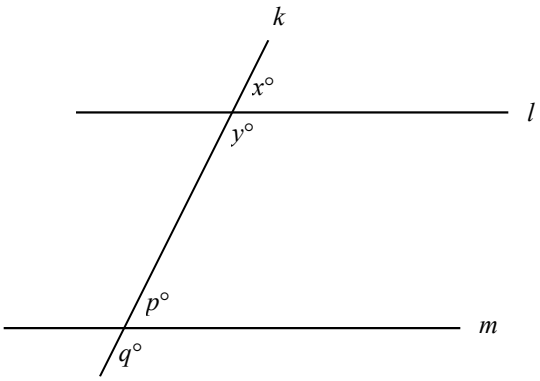
➢ **Easy**

1. Is xy a multiple of 48?

   (1) x is a multiple of 4.
   (2) y is a multiple of 6.

2. What is the last digit of the positive integer n ?

   (1) The last digit of n equals the last digit of n^2.
   (2) n is an even number.



3. In the figure, l, m, and k are straight lines. Are the lines l and m parallel?

   (1) $x = q$
   (2) $y = p$

4. What is the value of $l + 2t$?

   (1) $l + t = 4$
   (2) $l + 3t = 9$

5. If p is the circumference of the circle Q, what is the value of p ?

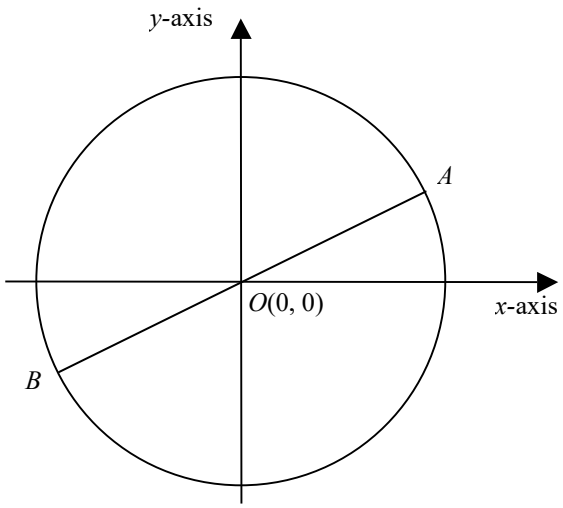
   (1) The radius of circle Q is 5.
   (2) The area of circle Q is 25π.

6. What is the value of x ?

   (1) x is the sum of all prime numbers greater than 49 and less than 24,632.
   (2) x is the sum of all odd prime numbers greater than 50 and less than 24,650.

7. In the figure shown, AB is a diameter of the circle and O is the center of the circle. What is the value of circumference of the circle?

   (1) $A = (3, 4)$
   (2) The diameter of the circle is 10.



➢ **Medium**

8. Is $x^2 > y^2$?

   (1) $x > y > 0$
   (2) $x > 1 > y$

9. Is $\frac{a^2 - b^2}{a - b} > \frac{a^2 - b^2}{a \square b}$?

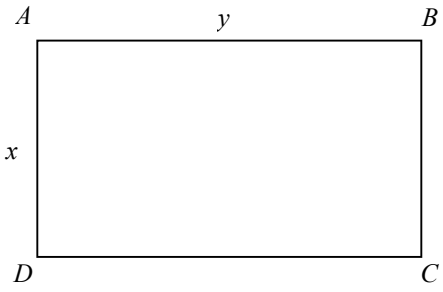
   (1) $a = 49$
   (2) $b = 59$

10. Is the number x a multiple of 12?

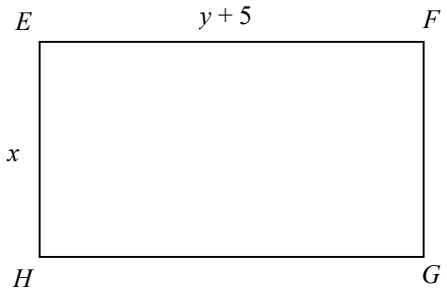
    (1) Both 3 and 4 divide into x evenly.
    (2) Both 2 and 6 divide into x evenly.

11. Station Q is to the East of Station T. At 12 noon, a train starts from Station Q and travels at a constant speed of x mph towards Station T. At 12 noon of the same day, another train starts from Station T and travels at a constant speed of y mph towards Station Q. At what time will the trains meet?

    (1) $y = 4x/3$
    (2) $x = 100$ mph



First Rectangle



Second Rectangle

12. In the figure, the area of the first rectangle is 100. What is the area of the second rectangle?

(1) $x = 5$
(2) $y = 20$

13. Is $N/2$ an odd integer?

(1) $2N$ is even.
(2) N is even.

14. Is $a/b > 0$?

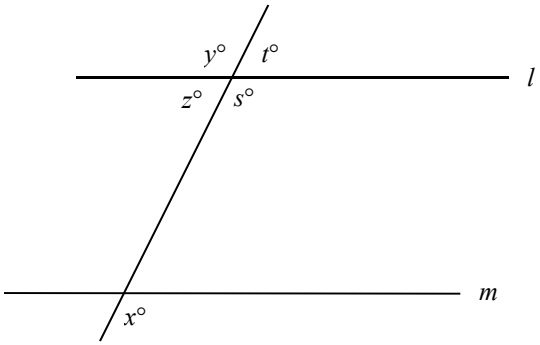
(1) $ab > 0$
(2) $a - b > 0$

15. Is $\sqrt{m \square n}$ an integer?

(1) $\sqrt{m}$ is an integer.
(2) $\sqrt{n}$ is an integer.

16. How much is 20 percent of a certain number a ?

(1) 20 is 20 percent of the number a.
(2) 80 percent of the number a is 80.



17. In the figure, lines l and m are parallel. What is the value of x ?

(1) $z + t = 120$
(2) $y - z = 60$

18. If r and s are two positive numbers, what is the value of the ratio r/s ?

(1) r is 25% greater than s.
(2) r is 25 units greater than s.

19. Is it true that $x - z > y - z$?

(1) $xz > yz$
(2) $x + z > y + z$

20. If $x - 4y = 1$, what is the value of x ?

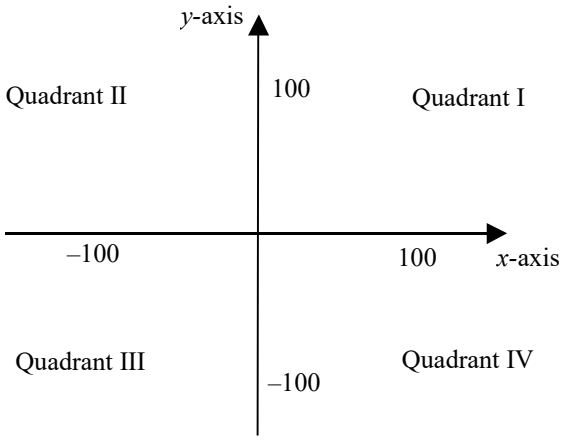
(1) $x = 4y + 1$
(2) $y = x/2 + 1$

21. Is $\frac{j}{m} \square \frac{k}{n} \square 2$?

(1) $\frac{jk}{mn} \square 1$
(2) $\frac{jn}{mk} \square 1$

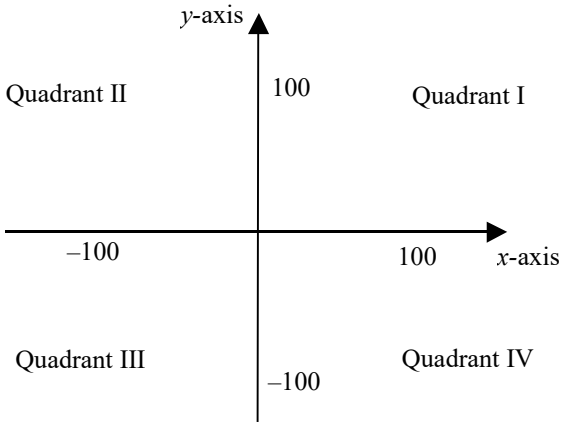
22. A, B, C, and D are points on a line in that order. If $AD = 13$, what is the length of BC ?

(1) $AC = 5$
(2) $BD = 10$



23. In which quadrant does the point (a, b) lie if $ab \neq 0$?

(1) (b, a) lies in Quadrant III.
(2) $a/b > 0$



24. Is $a + b > 0$?

(1) The point (a, b) is in Quadrant II.
(2) $|a| - |b| > 0$

25. Is $a - b > 2.5$?

(1) $a > b + 2.5$
(2) $a = 3 + b$

26. Is the positive integer x an odd number?

(1) $x^2 + 3x + 4$ is even.
(2) $x^2 + 4x + 3$ is odd.

27. Is the positive integer x an odd number?

(1) $x^2 + 3x + 3$ is odd.
(2) $|x| + x = 4$

28. Which one of the expressions x^2y and x^2/y is greater?

(1) $y > x$
(2) $y < -1$

29. What is the value of x ?

(1) $x - 3 = 10/x$
(2) $x > 0$

30. What is the value of $x + y$?

(1) $2x + 3y = 11$
(2) $3x + 2y = 9$

31. Is $x > y$?

(1) $2x = 2y + 1$
(2) $3x = 4y - 1$

32. Is x positive?

(1) $x + 3$ is positive.
(2) $x - 3$ is positive.

33. Is $2p - 3q$ not equal to 0?

(1) $4p$ is not equal to $6q$.
(2) p is not equal to q.

34. If m and n are positive integers, then is $m + 2n$ an even integer?

(1) $m = 3n + 1$
(2) Exactly one of the two numbers m and n is even.

35. What is the value of x ?

(1) $2x + 1 = 10$
(2) $(2x + 1)^2 = 100$

36. In a sequence, the nth term a_n is defined by the rule $(a_{n-1} - 3)^2$. What is the value of a_4 ?

(1) $a_1 = 1$
(2) $a_2 = 4$

37. What is the numerical difference between the 5th and 6th terms in the sequence of numbers $a_0, a_1, a_2, \ldots, a_n$?

(1) The mth term is defined by the rule $2m + 1$.
(2) The 6th term is 81.

38. Can x equal 0 ?

(1) $x^2 + 1 > 2x + 4$
(2) $(x + 1)^2 - 2x > 2(x + 1) + 2$

39. Is x greater than y ?

(1) $x = 2y$
(2) $y = 2k$

40. If $y = -x$, then is $y > -1$?

(1) $x^4 > x^2$
(2) $x^3 < x^2$

41. $2x - y =$

(1) $3x + y = x + 2y$
(2) $(x + y) + 3(x - y) = 0$

42. What is the value of $\frac{xy}{x \square y}$?

(1) $1/x + 1/y = 1/3$
(2) $x + y = xy/3$

43. What is the value of $(x - 2)^2$?

(1) $x^2 - 4x + 3 = 0$
(2) $x^2 - 7x + 12 = 0$

44. What number is 25 percent of x ?

(1) 50 percent of x is 50.
(2) 9/100 of x is 9.

45. What is the value of $x - y$?

(1) x is 50% more than y.
(2) x is the sum of y and 20.

46. What is the value of $x + y^2$?

(1) $x + y = 7$
(2) $x^2 + y^2 = 25$

47. Is $x/15 > y/25$?

(1) $5y + 3x > 8x + 2y$
(2) $6y + 5x > 10x + 3y$

48. Is $a > b$?

(1) The arithmetic mean of a and b is 25.
(2) The geometric mean of a and b is 25.

49. Is $a > b$?

(1) The arithmetic mean of a and b is 25.
(2) The geometric mean of a and b is 24.

50. What is the value of $x^2 - y^2$?

(1) $x + y = 12$
(2) $xy = 35$

51. What is the value of $\frac{a \square b}{\sqrt{ab}}$?

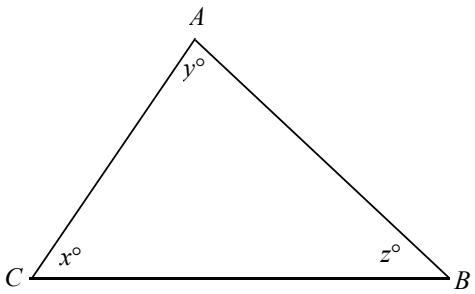
(1) a is 4 times b.
(2) b is one-fourth of a.

52. $xy - yz =$

(1) $yz - zx = 3$
(2) $zx - xy = 4$

53. Is $x > y$?

(1) $x^5 > y^5$
(2) $x^4 < y^4$



54. In figure, *ABC* is a triangle. What is the length of side *AB* ?

(1) $x = 50$
(2) $y = 45$

55. What is the minimum area of cardboard required to make a rectangular box with no top?

(1) The area of the bottom of the box is 23 square feet.
(2) The volume of the box is 23 cubic feet.

56. What is the value of $ab + bc$?

(1) $b = a + c$
(2) $b = 3$

57. If the value of $(x - y)(x + y)$ is 15, then what is the value of x/y ?

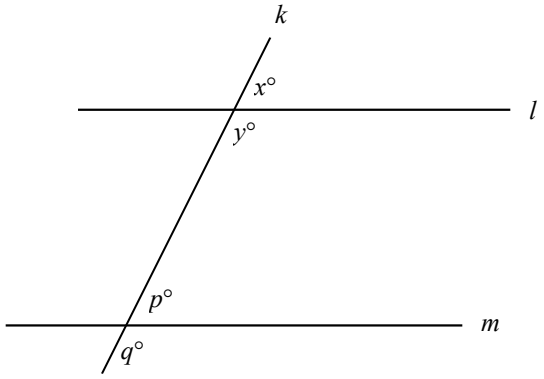
(1) $x - y = 3$
(2) $x + y = 5$

58. Steve deposited $100 to open a savings account. If there are no other transactions in the account, what amount of money would the account accrue in 6 months after opening the account?

(1) The interest rate is 4%.
(2) Interest is compounded quarterly.

59. $m =$

(1) $7m + 5n = 29$
(2) m and n are positive integers.



60. In the figure, *l*, *m*, and *k* are straight lines. Are the lines *l* and *m* parallel to each other?

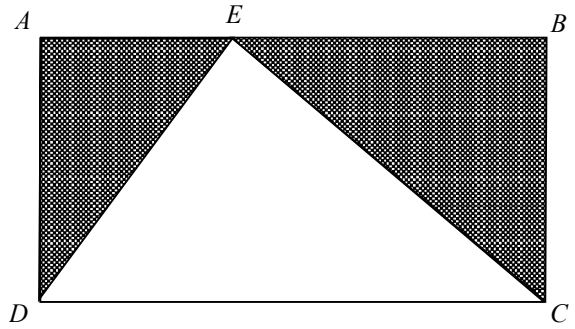
(1) $x = q$
(2) $y = q$

61. Is x divisible by 14?

(1) $x/4$ is a positive integer.
(2) $x/6$ is a positive integer.

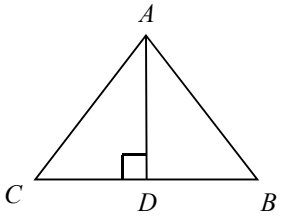
62. Waugh jogged to a restaurant at x miles per hour, and jogged back home along the same route at y miles per hour. He took 30 minutes for the whole trip. What is the average speed at which he jogged for the whole trip?

(1) The restaurant is 2 miles from home.
(2) $x = 12$ and $y = 6$.



63. In the figure, $ABCD$ is a rectangle and E is a point on the side AB. What is the area of the shaded region in the figure?

(1) $AB = 10$ and $AD = 5$
(2) $AE = 4$ and $EB = 6$



64. What is the area of the equilateral triangle shown in the figure?

(1) The base $BC = 6$.
(2) The altitude $AD = 3\sqrt{3}$.

65. If $x^2 - y^2 = 15$, is $x + y > x - y$?

(1) $x - y = 5$
(2) $x + y = 3$

66. Is $(x - 2)(x - 3) > 0$?

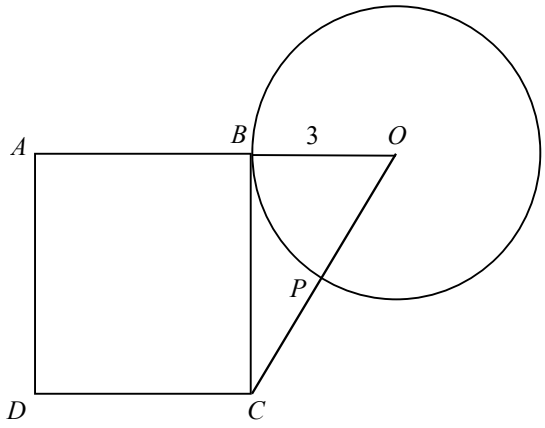
(1) $x > 2$
(2) $x < 3$

67. If $x > y$, then is $1/x < 1/y$?

(1) x is negative.
(2) y is negative.

68. A rectangular field is 3.2 yards long. What is the area of the field in square yards?

(1) A fence marking the boundary is 11.2 yards in length.
(2) The distance between the diagonally opposite corners of the rectangular field is 4 yards.



69. In the figure, *ABCD* is a square, and *OB* is a radius of the circle. Is *BC* a tangent to the circle?

(1) $PC = 2$
(2) Area of the square is 16.

70. Is the product of two numbers x and y a prime number?

(1) x and y are two prime numbers.
(2) x and y are two odd numbers not equal to 1.

71. Is $(x - 2)(x - 3) > 0$?

(1) $x < 2$
(2) $x < 3$

72. What is the average score of the test takers on a test?

(1) Exactly half the test takers scored above 500.
(2) The lowest and the highest test scores are 200 and 800, respectively.

73. What is the value of $2p + q$?

(1) $3p + 2q = 5$
(2) $6p + 4q = 10$

74. Is $x > y$?

(1) $x = y + 1$
(2) $x + y = 1$

75. If x equals $y/7$, then is x less than 30?

(1) $y > 210$
(2) $y < 217$

76. Is $\frac{|a-b|\cdot|c-b|}{d} \neq 0$?

(1) $a + c < 2b$
(2) $d < 0$

77. Does the integer m have at least 4 positive prime factors?

(1) $m/14$ is a positive integer.
(2) $m/15$ is a positive integer.

78. Does the positive integer n have at least 3 positive prime factors?

(1) $n/10$ is an integer.
(2) $n/30$ is an integer.

79. If x is not equal to 1, then is y equal to zero?

(1) $y = \frac{1}{x-1}$
(2) $y = \frac{x}{6}$

80. The savings from a person's income is the difference between his or her income and expenditure. What is the ratio of Mr. Kelvin's savings in the year 1998 to the year 1999?

(1) The ratio of his income in the years 1998 to 1999 is 3 : 4.
(2) The ratio of his expenditure in the years 1998 to 1999 is 5 : 6.

81. What is the average temperature from Monday through Saturday of a week?

(1) The average temperature from Monday through Wednesday of the week is 36°C.
(2) The minimum and maximum temperatures between Thursday and Saturday of the week are 25°C and 38°C, respectively.

82. If $a < b$, is $(a - b) + (c - b)$ positive?

(1) $c < a$
(2) $c < b$

➢ **Hard**

83. $\frac{1}{p^4} = \frac{1}{q^4}$

(1) $p + q = 12$
(2) $pq = 35$

84. If $H = \frac{T\sqrt{1-R^2}}{k\sqrt{1-g^k}}$, then $H =$

(1) $T = 3$, $R = 1/2$
(2) $k = 4$, $g = 1/3$

85. If $H = 3$, then is $\sqrt{H^2 - 2H}$ equal to $\frac{T\sqrt{1-R^2}}{k\sqrt{1-g^k}}$?

(1) $T = 3$, $R = 1/2$
(2) $k = 1$, $g = 7/16$

86. What fraction of the students in a class have blue eyes?

(1) 10% of the girls have blue eyes.
(2) 20% of the boys have blue eyes.

87. If n is a positive integer, then is $(n + 1)(n + 3)$ a multiple of 4?

(1) $(n + 2)(n + 4)$ is odd.
(2) $(n + 3)(n + 6)$ is even.

88. X is a 4-digit integer, and Y is a 3-digit integer. Is the sum of the digits of X greater than the sum of the digits of Y ?

(1) All the digits of X are greater than 5.
(2) All the digits of Y are less than 5.

89. In a factory, there are workers, executives, and clerks. How many employees are there in the factory?

(1) 41% of the employees are workers, 460 are executives, and the remaining 720 employees are clerks.
(2) 460 of the employees are executives and account for 23% of the employees in the factory.

90. How many families in the town of Windsor have exactly one car?

(1) 250 families have at least one car.
(2) 60 families have at least two cars.

91. The population of the town of Paxton continuously increased in the two decades 1960 through 1969 and 1970 through 1979. Was the percentage increase greater in the first decade than in the second decade?

(1) The population increase in the first decade was 10,082.
(2) The population increased by the same amount in both decades.

92. The length and width of a rectangular box are 6 feet and 5 feet, respectively. A hose supplies water at a rate of 6 cubic feet per minute. How long would it take to fill a conical box whose volume is three times the volume of the rectangle box?

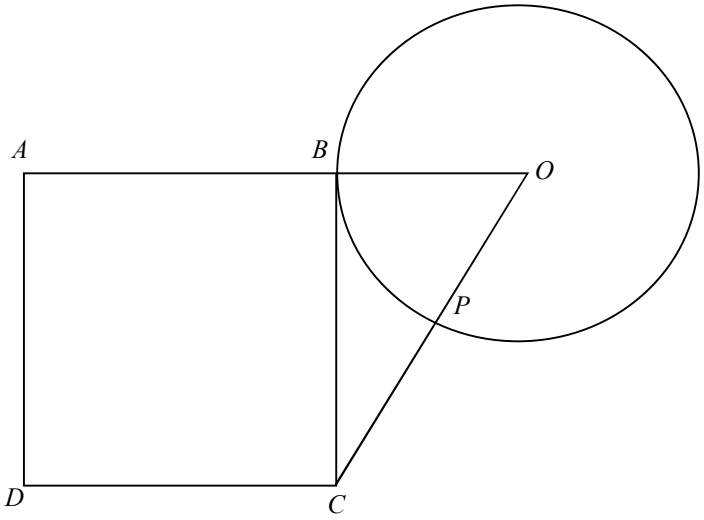
(1) The depth of the rectangular box is 7 feet.
(2) The radius of the base of the conical box is 6 feet.

93. If a, b, and c are three different numbers, then what is the value of $ax + by + cz$?

(1) $x/(b - c) = y/(c - a) = z/(a - b)$
(2) $ax : by : cz = 1 : 2 : -3$

94. Is ΔABC a right triangle?

(1) The angles of ΔABC are in the ratio 1 : 2 : 3.
(2) One of the angles of ΔABC equals the sum of the other two angles.



95. In the figure, O is the center of the circle of radius 3, and $ABCD$ is a square. If $\angle AOP = x$ and the side BC of the square is tangent to the circle, then what is the area of the square $ABCD$?

(1) $PC = 3$
(2) $x = 60$

96. Are the two triangles similar?

(1) The ratio of the measures of the angles in one triangle is 2 : 3 : 4.
(2) The ratio of the measures of the angles in the other triangle is 4 : 3 : 2.

97. Are the two triangles similar?

(1) The ratio of the measures of the angles in one triangle is 2 : 3 : 4.
(2) The ratio of the measures of the angles in the other triangle is 1 : 1 : 1.

98. What is the area of quadrilateral $ABCD$?

(1) $AB = CD = 3$
(2) $AD = BC = 4$

99. Is x greater than zero?

(1) $x^6 > x^2$
(2) $x^5 < x^2$

100. What is the height of a rectangular tank?

(1) The area of the base of the tank is 100 sq. ft.
(2) It takes 20 seconds to fill up the tank with water poured at the rate of 25 cubic feet per second.

101. What is the value of $x^3 + y^3$?

(1) $x + y = 7$
(2) $x^2 + y^2 = 25$

102. If x equals one of the numbers 1/54, 1/23, or 1/12, then

(1) $1/55 < x < 1/22$
(2) $1/33 < x < 1/11$

103. In ΔPQR, if $PQ = x$, $QR = x + 3$, and $PR = y$, which one of the three angles of ΔPQR is the greatest?

(1) $x > y$
(2) $x = y + 3$

104. What is the average speed of the car during its entire trip?

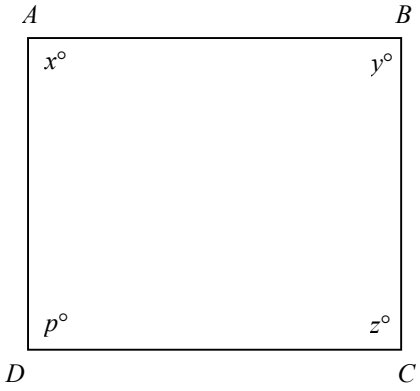
(1) The car traveled at 75 mph for the first half (by time) of the trip and at 40 mph for the second half of the trip.
(2) The car would have taken 5 hrs to complete the trip if it traveled at 75 mph for the entire trip.

105. If s and t are positive integers, then what is the remainder when s is divided by t ?

(1) $s/t = 39.13$
(2) $s = 17{,}731$ and $t = 429$

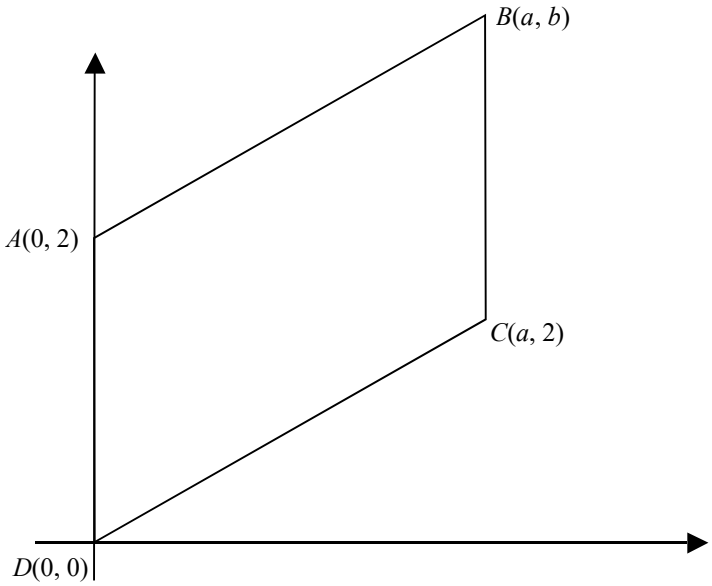
106. The cost of production of a certain instrument increases with the number of units produced. What is the cost of production for 270 units?

(1) The cost of production for 300 units is \$300.
(2) The cost of production is directly proportional to the number of units produced.



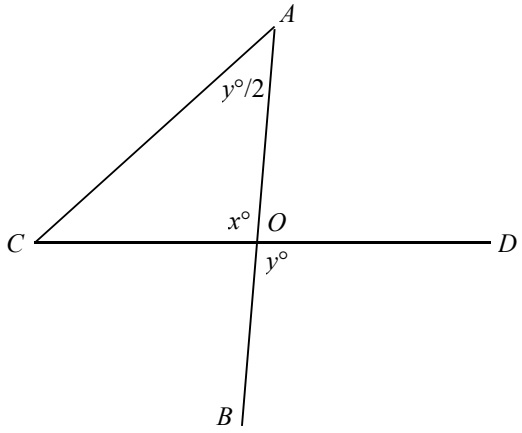
107. In the figure, is the quadrilateral $ABCD$ a square?

(1) $x = y = z = p$
(2) $AB = BC = CD = DA$



108. In the rectangular coordinate system shown, $ABCD$ is a quadrilateral. The coordinates of the points A, B, C, and D are (0, 2), (a, b), (a, 2), and (0, 0), respectively. Is the quadrilateral $ABCD$ a parallelogram?

(1) $a = 3$
(2) $b = 5$



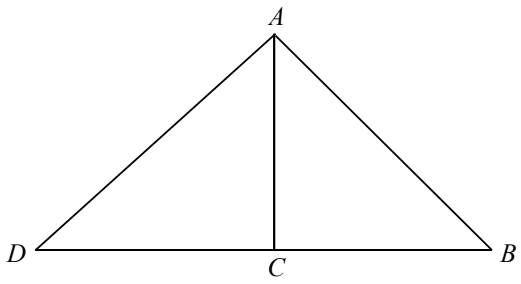
The dimensions in the figure may be different from what they appear to be.

109. In the figure, is the line AB perpendicular to the line CD?

(1) $x + y = 180$
(2) AOC is an isosceles triangle

110. A regular hexagon is inscribed in a circle. What is the area of the hexagon?

(1) The radius of the circle is 10.
(2) The perimeter of the hexagon is $40\sqrt{3}$.



The dimensions in the figure may be different from what they appear to be.

111. Is triangle ABC a right triangle?

(1) ACD is a right triangle.
(2) AC is the greatest side of ΔACD.

112. Is $x > y$?

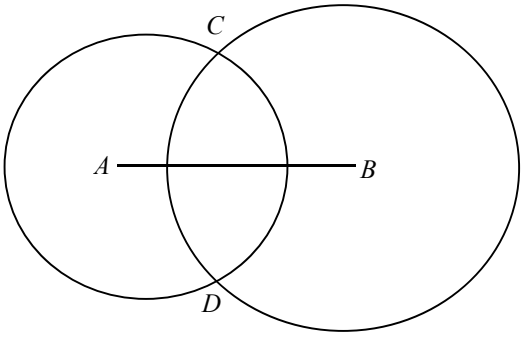
(1) $(x-y)^3 > (x-y)^2$
(2) $(x-y)^4 > (x-y)^3$

113. A worker is hired for 7 days. Each day, he is paid 11 dollars more than what he is paid for the preceding day of work. How much was he paid in total for the 7 days of work?

(1) The total amount he was paid in the first 4 days of work equaled the total amount he was paid in the last 3 days.
(2) The total amount he was paid in the last 3 days of work equaled \$66 more than the total amount he was paid in the first 3 days of work.

114. What is the sum of the two numbers x and y?

(1) The ratio of the sum of the reciprocals of x and y to the product of the reciprocals of x and y is 1 : 3.
(2) The product of x and y is 11/36 units greater than the sum of x and y.



115. In the figure, A and B are the centers of the two circles. The circles intersect each other at C and D. What is the value of the common area of the two circles?

(1) The radius of the circle with center A is 5.
(2) $\angle CAD = 120°$.

116. Is the product of the three nonzero numbers a, b, and c divisible by 81?

(1) The product of the three numbers a, b, and c is a multiple of 27.
(2) None of the three numbers a, b, and c is divisible by 9.

Answers and Solutions to Problem Set

➢ **Easy**

1. Not every multiple of 4 is a multiple of 48 (For example, 8 [= 4 · 2] is a multiple of 4, but 8 is not a multiple of 48). Hence, Statement (1) alone is not sufficient to answer the question.

Not every multiple of 6 is a multiple of 48 (For example, 12 [= 6 · 2] is a multiple of 6, but 12 is not a multiple of 48). Hence, Statement (2) alone is not sufficient to answer the question.

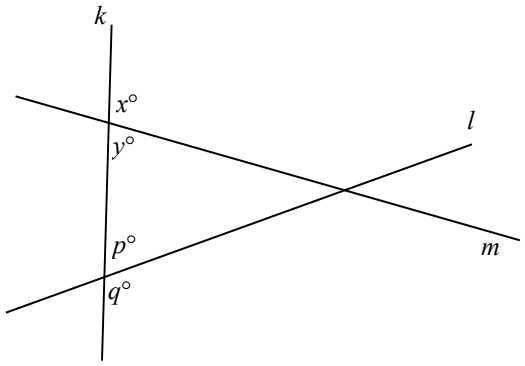
A multiple of both 4 and 6 must be a multiple of the least common multiple of 4 and 6, which is 24. But again, not every multiple of 24 is a multiple of 48 (For example, 72 [= 24 · 3] is a multiple of 24, but 72 is not a multiple of 48). Hence, statements (1) and (2) together cannot answer the question. The answer is (E).

2. The last digit of a positive integer is the same as the last digit of n^2 only when the last digit is 0, 1, 5, or 6. For example, The last digit of 10 is 0 and the last digit of 10^2 (= 100) is also 0. The last digit of 11 is 1 and the last digit of 11^2 (= 121) is also 1. The last digit of 15 is 5, and the last digit of 15^2 (= 225) is also 5. The last digit of 16 is 6, and the last digit of 16^2 (= 256) is also 6. Since we have multiple cases with Statement (1) alone, it is *not* sufficient.

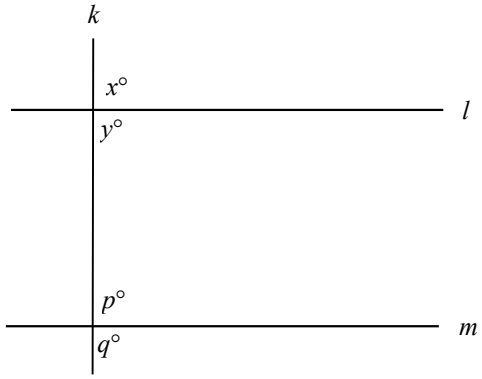
Now, the last digit of an even number is 0, 2, 4, 6, or 8. Hence, Statement (2) alone is *not* sufficient.

The common solutions from the two statements are 0 and 6 (which means n ends with either 0 or 6). Since we do not have a single solution, the statements together are not sufficient. The answer is (E).

3. In the figure below, statements (1) $x = q$ and (2) $y = p$ are satisfied. Still, lines l and m intersect and therefore are not parallel.



Now, consider the figure below, where each of the angles x, y, p, and q equals 90°. Here, the statements are satisfied and lines l and m are parallel.



Since we have a double case, the statements together are *not* sufficient. Hence, the answer is (E).

4. Statement (1) contains two variables and only one equation. Hence, it cannot determine the value of $l + 2t$. So, Statement (1) alone is *not* sufficient.

Similarly, Statement (2) contains two variables and only one equation. Hence, it cannot determine the value of $l + 2t$. So, Statement (2) alone is *not* sufficient.

With Statement (1) and Statement (2) together, we have two equations in two unknowns:

$$l + t = 4$$
$$l + 3t = 9$$

This system may have a solution. To find out, add the equations:

$$(l + t) + (l + 3t) = 4 + 9$$

$$2l + 4t = 13$$

$$l + 2t = 13/2$$

Hence, the statements together answer the question. The answer is (C).

5. If we know the radius of the circle, we can draw a unique circle with the radius and can then measure the circumference. So, Statement (1) alone is sufficient.

Now, the area of the circle $Q = \pi \times radius^2 = 25\pi$. Solving the equation for the radius yields $radius = \sqrt{\frac{25\pi}{\pi}} = 5$. Hence, we have the radius from Statement (2) alone, and therefore Statement (2) alone is sufficient. The answer is (D).

6. The number of primes greater than 49 and less than 24,632 is fixed and therefore can be summed and the question can be answered. Hence, Statement (1) alone is sufficient.

The number of odd primes that are greater than 50 and less than 24,650 is fixed and therefore can be summed and the question can be answered. Hence, Statement (2) alone is sufficient.

Since either Statement alone is sufficient, the answer is (D). This question merely tests your understanding of the Data Sufficiency format.

7. The formula for the circumference of a circle is $2\pi \times radius$. Hence, if we know the radius, then the circumference of the circle can be calculated.

The point $A(3, 4)$ is on the circle, and $O(0, 0)$ is the center of the circle. Hence, AO is a *radius*. Since the points can be located (because we have the coordinates of the two end points of the line segment), the distance between them AO (= *radius*) can be calculated and therefore Statement (1) alone is also sufficient.

The radius is half the diameter. The diameter is given in Statement (2), so the statement is sufficient.

The answer is (D).

➢ **Medium**

8. From the inequality $x > y > 0$ in Statement (1), we have that both x and y are positive. Multiplying the inequality $x > y$ by x yields $x^2 > xy$, and multiplying by y yields $xy > y^2$. Combining the two results yields $x^2 > xy > y^2$. So, we have $x^2 > y^2$. Hence, Statement (1) alone is sufficient to answer the question.

From Statement (2) alone, we have $x > 1 > y$. Now, if $x = 2$ and $y = -3$, then x^2 is not greater than y^2. But, if $x = 2$ and $y = -1$, then x^2 is greater than y^2. Hence, we have a double case and Statement (2) alone is *not* sufficient.

The answer is (A).

9. Applying the Difference of Squares formula to the numerator of $\frac{a^2 - b^2}{a - b}$ yields $\frac{(a + b)(a - b)}{a - b} = a + b$, and doing the same to $\frac{a^2 - b^2}{a + b}$ yields $\frac{(a + b)(a - b)}{a + b} = a - b$. So, the question reduces to "Is $a + b > a - b$?" Adding $-a + b$ to both sides of the inequality further reduces the question to "Is $2b > 0$?", or finally "Is $b > 0$?" Statement (2): $b = 59$ alone can answer the question. Statement (1): $a = 49$ alone is not relevant since it does not tell us the value of b. The answer is (B).

10. From Statement (1), we have x is divisible by 3 and 4. So, x must be a common multiple of 3 and 4. The least common multiple of 3 and 4 is 12. So, x is a multiple of 12. Hence, Statement (1) answers the question, "Is x multiple of 12?"

From Statement (2), we have 2 and 6 divide x evenly. The LCM of 2 and 6 is 6, so x is a multiple of 6. But, not all multiples of 6 are multiples of 12 (for example, 18). Hence, Statement (2) alone is not sufficient to answer the question, "Is x is multiple of 12?"

The answer is (A).

11. The time at which the trains meet depends on the distance between the stations. So, from the speeds alone, we cannot evaluate the time taken. Hence, statements (1) and (2) together are not sufficient to answer the question. The answer is (E).

12. The area of a rectangle is *length* × *width*. Hence, the area of the first rectangle is $xy = 100$ (given that the area is 100).

The area of the second rectangle is $x(y + 5) = xy + 5x = 100 + 5x$ (since $xy = 100$). Hence, the value of x is sufficient to answer the question. Hence, Statement (1) alone is sufficient.

Now, if we have the value of y, we can evaluate x as follows: Since $xy = 100$, $x = 100/y$. Hence, Statement (2) alone is sufficient.

Since either statement is sufficient, the answer is (D).

13. Suppose $N/2$ is odd. Then $N = 2 \times$ (an odd number) = even. Also, $2N = 2 \times$ (even number) = even. Hence, both statements (1) and (2) are satisfied.

If $N/2$ is even, then $N = 2 \times$ (an even number) = even, and $2N = 2 \times$ (an even number) = an even number. Here, both statements (1) and (2) are satisfied.

Hence, the two statements are valid both when $N/2$ is odd and when $N/2$ is even. Hence, the statements together are not sufficient to answer the question. The answer is (E).

14. Dividing the inequality $ab > 0$ by b^2, a positive value, yields $a/b > 0$. Hence, Statement (1) alone is sufficient.

Now, $a - b$ is positive when a is greater than b. When both a and b are positive, a/b is positive. When a is positive and b is negative, a/b is negative. Hence, we have a double case, and Statement (2) alone is not sufficient.

Hence, the answer is (A): Statement (1) alone is sufficient, and Statement (2) alone is not sufficient.

15. Let $m = 9$ and $n = 16$. Then $\sqrt{m} = 3$ is an integer, and $\sqrt{n} = 4$ is also an integer. Now, $\sqrt{m \times n} = \sqrt{19 \times 16} = \sqrt{25} = 5$ is also an integer.

Now, let $m = 4$ and $n = 9$. Then $\sqrt{m} = 2$ is an integer, and $\sqrt{n} = 3$ is also an integer. But here, $\sqrt{m \times n} = \sqrt{4 \times 9} = \sqrt{13}$, not an integer.

Hence, we have a double case. The answer is (E): the two choices together are insufficient to answer the question.

16. From Statement (1) alone, we have that 20 is 20 percent of a. Hence, we have $20/100 \times a = 20$. Solving this equation for a yields $a = 100/20 \times 20 = 100$. Hence, Statement (1) alone is sufficient.

Now, from Statement (2) alone, we have that 80 is 80 percent of a. Now, 80 percent of a is $80/100 \times a$. Equating this to 80 yields $80/100 \times a = 80$. Solving this equation for a yields $a = 100/80 \times 80 = 100$. Hence, Statement (2) alone is sufficient.

The answer is (D).

17. Equating vertical angles in the figure yields $y = s$ and $z = t$.

Since lines l and m are parallel, the corresponding angles must be equal. So, $x = s$. Hence, the question is equivalent to "What is the value of s?"

Also, since a line has 180°, summing the angles z and s, which make the line l, yields $z + s = 180$. Solving for s yields $s = 180 - z$. From this equation, s is a dependent variable that depends only on z. Hence, if we know the value of z, we can determine the value of s, which answers the question. Hence, the question is equivalent to "What is the value of z?"

Now, from Statement (1) alone, we have that $z + t = 120$. Since $z = t$, we have $z + z = 120$, or $2z = 120$, or $z = 120/2 = 60$. Hence, Statement (1) alone determines z.

Now, from Statement (2) alone, we have that $y - z = 60$. Adding z to both sides yields $y = z + 60$. Now, a straight angle has 180°. Hence, we have

$$y + z = 180$$
$$(z + 60) + z = 180$$
$$2z = 180 - 60 = 120$$
$$z = 120/2 = 60$$

Hence, Statement (2) alone also determines z.

Hence, the answer is (D).

18. From Statement (1) alone, we have that r is 25% greater than s. Hence, $r = (1 + 25/100)s$. Hence, $r/s = 1 + 25/100 = 1.25$. Hence, Statement (1) alone is sufficient.

From Statement (2) alone, we have that r is 25 units greater than s. Hence, we have the equation $r = s + 25$. Dividing the equation by s yields $r/s = s/s + 25/s = 1 + 25/s$. Since s is unknown, $1 + 25/s$ cannot be calculated. Hence, Statement (2) alone is not sufficient.

Hence, the answer is (A).

19. Canceling z from both sides of the inequality $x - z > y - z$ yields $x > y$. Hence, $x - z$ is greater than $y - z$ when $x > y$. So, the question reduces to "Is $x > y$?"

From Statement (1) alone, we have the inequality $xz > yz$.

Suppose z is positive. Then we can divide the inequality $xz > yz$ by z without flipping the direction of the inequality, and we get $x > y$.

Suppose z is negative. Then we can divide the inequality $xz > yz$ by z while flipping the direction of inequality, and we get $x < y$.

We have a double case, and therefore Statement (1) alone is not sufficient.

From Statement (2) alone, we have that $x + z > y + z$. Subtracting z from both sides yields $x > y$. Hence, Statement (2) alone is sufficient to answer the question.

The answer is (B).

20. We are given the linear equation $x - 4y = 1$. Adding $4y$ to both sides yields $x = 4y + 1$. So far, we have only one equation in two unknowns.

From Statement (1) alone, we have the equation $x = 4y + 1$, which is same as the given equation. Hence, effectively, we still have only one equation in two unknowns, x and y. So, Statement (1) alone is not sufficient to determine the value of x.

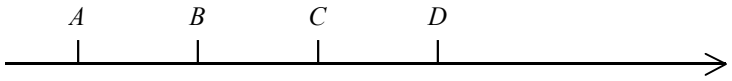
From Statement (2) alone, we have that $y = x/2 + 1$. This is a different linear equation in the same two unknowns. Hence, together with the given equation $x = 4y + 1$, we are able to determine a unique value for x. The answer is (B). Solving the equations yields $x = –5$ and $y = –3/2$.

21. From Statement (1) alone, we have that $jk/mn = 1$. Solving for j yields $j = mn/k$. Hence, the expression $\frac{j}{m} \cdot \frac{k}{n}$ equals $\frac{mn/k}{m} \cdot \frac{k}{n} \cdot \frac{n}{k} \cdot \frac{k}{n}$. Since we still do not have the ratio k/n, Statement (1) alone is not sufficient.

From Statement (2) alone, we have that $jn/mk = 1$. Solving for j yields $j = mk/n$. Hence, the expression $\frac{j}{m} \cdot \frac{k}{n} n$ equals $\frac{mk/n}{m} \cdot \frac{k}{n} \cdot \frac{k}{n} \cdot \frac{k}{n} \cdot \frac{2k}{n}$. Since we still do not have the ratio k/n, Statement (2) alone is also not sufficient.

Now, equating the results of the expression from the two statements yields $\frac{n}{k} \cdot \frac{k}{n} \cdot \frac{2k}{n}$. Subtracting $\frac{k}{n}$ from both sides yields $\frac{n}{k} \cdot \frac{k}{n}$. Multiplying both sides by $\frac{k}{n}$ yields $\left(\frac{k}{n}\right)^2 \cdot 1$; $\frac{k}{n} = +1$ or $–1$. Hence, $\frac{2k}{n}$ equals $2(–1) = –2$ or $2(1) = 2$. Hence, we have a double case, and even the statements together do not answer the question. Hence, the answer is (E).

22. Drawing a number line with the points A, B, C, and D yields



From the number line, we have the equations

$$AC = AB + BC$$
$$BD = BC + CD$$
$$AD = AB + BC + CD$$

Adding the first two equations yields

$$AC + BD = (AB + BC) + (BC + CD) = AB + 2BC + CD$$

Subtracting the third equation from this equation yields

$$\begin{aligned} AC + BD - AD &= AB + 2BC + CD - (AB + BC + CD) \\ &= BC \end{aligned}$$

Hence, $BC = AC + BD - AD = AC + BD - 13$. Now, the side lengths of BD and AC are independent of each other. Hence, the value of BC depends on both AC and BC. Hence, both statements are required to answer the question. Hence, the answer is (C), and $BC = AC + BD - 13 = 5 + 10 - 13 = 2$.

23. From Statement (1) alone, we have that the point (b, a) lies in the third quadrant. In the third quadrant, both x- and y-coordinates are negative. So, b and a are negative and therefore (a, b) also lies in the same quadrant (Quadrant III). Hence, Statement (1) alone is sufficient.

From Statement (2) alone, we have that a/b is greater than 0 (positive). This happens when both a and b are positive or when both a and b are negative. In the first case, (a, b) is in Quadrant I; and in the second case, (a, b) is in Quadrant III. So, we have a double case, and Statement (2) alone is not sufficient.

The answer is (A).

24. From Statement (1) alone, we have that the point (a, b) is in Quadrant II. Hence, a is negative and b is positive. Since $|a|$ is the positive value of a and a is negative, a can be expressed as $-|a|$. Also, $|b|$ is the positive value of b; and since b is positive, b can be expressed as $|b|$. Hence, $a + b = -|a| + |b|$. Now, if $|b|$ is greater than $|a|$, $a + b$ is greater than 0. Otherwise, $a + b$ is not greater than 0. Hence, we have a double case, and Statement (1) alone is not sufficient.

From Statement (2) alone, we have that $|a| - |b| > 0$. Adding $|b|$ to both sides yields $|a| > |b|$. Now, suppose a equals 3 and b equals 2. Then $a + b = 5$, a positive number. Now, suppose a is -3 and b is 2. Then $a + b = -1$, a negative number. Hence, we have a double case, and even Statement (2) alone is not sufficient.

Combining the results from the two statements, $a + b = -|a| + |b|$ and $-|a| + |b| < 0$, yields $a + b < 0$. Hence, the statements together determine that $a + b$ is not greater than 0. Hence, the answer is (C).

25. From Statement (1) alone, we have the inequality $a > b + 2.5$. Subtracting b from both sides yields $a - b > 2.5$. Hence, Statement (1) alone is sufficient to answer the question.

From Statement (2) alone, we have that $a = 3 + b$. Subtracting b from both sides yields $a - b = 3$. Since 3 is greater than 2.5, $a - b$ is greater than 2.5. Hence, Statement (2) alone is also sufficient to answer the question.

The answer is (D).

26. From Statement (1) alone, we have that $x^2 + 3x + 4$ is an even number. Hence, $x^2 + 3x$ = (an even number) – 4 = an even number. So, $x^2 + 3x = x(x + 3)$ is an even number. The product of the two positive integers is even when at least one of them is even. So, whether x is even or $x + 3$ is even (because x is odd), $x(x + 3)$ is even. So, we cannot determine whether x is even or odd. Therefore, Statement (1) alone is not sufficient.

From Statement (2) alone, we have that $x^2 + 4x + 3$ is an odd number. Hence, $x^2 + 4x$ = (an odd number) – 3 = an even number. So, $x^2 + 4x = x(x + 4)$ is an even number. The product of two positive integers is even when at least one of them is even. So, at least one of the two factors x and $x + 4$ must be even. But, if x is even, $x + 4$ is even; and if $x + 4$ is even, x is even. Hence, Statement (2) alone is sufficient. The answer to the question is "No. x is even."

So, the answer is (B): Statement (2) alone is sufficient, and Statement (1) is not sufficient.

27. From Statement (1) alone, we have that $x^2 + 3x + 3$ is an odd number. Hence, $x^2 + 3x$ = (an odd number) – 3 = an even number. So, $x^2 + 3x$, which equals $x(x + 3)$, must be even. The product of the two positive integers x and $x + 3$ is even when at least one of them is even. Now, if $x + 3$ is even, x is odd; and if $x + 3$ is odd, x is even. So, x may or may not be odd. Statement (1) alone is not sufficient.

From Statement (2) alone, we have that $|x| + x = 4$. For positive numbers, $|x|$ equals x. Hence, $x + x = 4$; $2x = 4$; $x = 4/2 = 2$, an even number. Hence, Statement (2) alone is sufficient. The answer to the question is "No. x is even."

The answer is (B).

28. We can reword the question as either "Is $x^2y > x^2/y$?" or "Is $x^2y < x^2/y$?" A valid assumption is $x^2y \neq x^2/y$ since the original question asks for the greater one of the two. Let's work on "Is $x^2y > x^2/y$?" Dividing this inequality by the positive value x^2 changes the question to "Is $y > 1/y$?" Since the inequality in the question no longer has x^2, any further information about x is irrelevant. So, Statement (1) is *not* required.

From Statement (2) alone, we have that $y < –1$. Hence, y is negative. Dividing both sides of this inequality by $–y$, a positive number, yields $–1 < 1/y$. Combining the inequalities $y < –1$ and $–1 < 1/y$ yields $y < –1 < 1/y$ from which we have $y < 1/y$. Hence, Statement (2) alone answers the question. Hence, the answer is (B). The answer to the question is "No. y is not greater than $1/y$." So, the answer to the original question is "No. x^2y is *not* greater than x^2/y," or "x^2/y is the greater one."

29. From Statement (1) alone, we have the equation $x – 3 = 10/x$. Multiplying the equation by x yields $x^2 – 3x = 10$. Subtracting 10 from both sides yields $x^2 – 3x – 10 = 0$. Factoring the equation yields $(x – 5)(x + 2) = 0$. Now, the left-hand side of the equation is 0 when x equals 5 or –2. Since we have two solutions for x, Statement (1) alone is not sufficient.

From Statement (2) alone, we have a range of values for x ($x > 0$, x is positive), not a single value. Hence, Statement (2) alone is not sufficient to answer the question.

Combining the results from the two statements, we eliminate the chance of x being equal to –2 since now we know that x is positive. So, we are left with only one solution $x = 5$. Hence, the answer is (C).

30. Let $x + y = t$. Then $y = t - x$. Substituting this in the question changes it to

"What is the value of t?"

(1) $2x + 3(t - x) = 11$ or $3t - x = 11$
(2) $3x + 2(t - x) = 9$ or $2t + x = 9$

From Statement (1) alone, we have $3t - x = 11$. Solving the equation for t yields $t = (11 + x)/3$. Hence, the value of t is dependent on x (which is still unknown). Hence, Statement (1) alone is not sufficient to evaluate t.

Similarly, from Statement (2) alone, we have $2t + x = 9$. Solving the equation for t yields $t = (9 - x)/2$. Hence, the value of t is dependent on x (which is still unknown). Hence, Statement (2) alone is not sufficient to evaluate t.

Next, from statements (1) and (2) together, we have $3t - x = 11$ and $2t + x = 9$. Summing the two equations yields $5t = 20$; $t = 20/5 = 4$. Hence, the statements together answer the question. The answer is (C).

31. Dividing both sides of the equation in Statement (1) by 2 yields $x = y + 1/2$. Reading the equation yields "x is 1/2 unit greater than y." Hence, Statement (1) alone is sufficient.

Subtracting y from both sides of the inequality yields $x - y > 0$. Hence, the question can be changed to "Is $x - y > 0$?" or "Is $x - y$ positive?"

Let's derive an expression for $x - y$ from Statement (2) alone. The equation in the statement is $3x = 4y - 1$. Subtracting $3y$ from both sides of the equation yields $3x - 3y = y - 1$. Dividing both sides by 3 yields $x - y = (y - 1)/3$. Now, the value of $(y - 1)/3$ is less than 0 (when $y = -1$) and greater than 0 (when $y = 2$). Hence, Statement (2) alone is not sufficient to answer the question.

Hence, the answer is (A).

32. The question is "Is x positive?" In other words, "Is $x > 0$?"

From Statement (1) alone, we have that $x + 3$ is positive. Expressing this as an inequality yields $x + 3 > 0$. Subtracting 3 from both sides yields $x > -3$. Hence, x may have a negative value, such as -2 or -0.5, or a positive value, such as 1 or 2. Hence, we have a double case, and Statement (1) alone is not sufficient.

From Statement (2) alone, we have that $x - 3$ is positive. Expressing this as an inequality yields $x - 3 > 0$. Adding 3 to both sides of the inequality yields $x > 3$. Hence, x is positive. Hence, Statement (2) alone is sufficient to answer the question. The answer is "Yes. $x > 0$ is true."

Hence, the answer is (B).

33. From Statement (1) alone, we have that $4p \neq 6q$. Dividing both sides by 2 yields $2p \neq 3q$. Subtracting $3q$ from both sides yields $2p - 3q \neq 0$. Hence, Statement (1) alone is sufficient to answer the question.

From Statement (2) alone, we have $p \neq q$. Multiplying both sides by 2 yields $2p \neq 2q$. Subtracting $3q$ from both sides yields $2p - 3q \neq -q$. Now, if q equals 0, then $2p - 3q$ does not equal 0. But if q does not equal 0, then $2p - 3q$ may equal 0 [For example, when, say, $p = 3$ and $q = 2$, then $2p - 3q = 2(3) - 3(2) = 0$]. Hence, we have a double case, and Statement (2) alone is not sufficient.

Since Statement (1) alone is sufficient, the answer is (A).

34. From Statement (1) alone, we have $m = 3n + 1$. Hence, $m + 2n$ equals $(3n + 1) + 2n = 5n + 1$.

If n is even, then $5n$ is even and $5n + 1$ is odd.

If n is odd, then $5n$ is odd and $5n + 1$ is even.

Hence, we have a double case, and Statement (1) alone is not sufficient. Also, in either case, only one of the two numbers m and n is even. Hence, Statement (2) is a derivation of Statement (1). Hence, we can use the same two cases in Statement (1) to determine that Statement (2) alone is not sufficient.

Also, since Statement (2) is a derivation of Statement (1), the same two cases in Statement (1) can be used to determine that the statements together are not sufficient to answer the question. Hence, the answer is (E).

35. Solving the equation $2x + 1 = 10$ in Statement (1) for x yields $x = 9/2$. Hence, Statement (1) alone is sufficient.

From Statement (2) alone, we have that $(2x + 1)^2 = 100$. Square rooting both sides yields two equations: $2x + 1 = 10$ and $2x + 1 = –10$. Solving the first equation for x yields $x = 9/2$, and solving the second equation for x yields $x = –11/2$. Hence, we have a double case, and Statement (2) alone is not sufficient to answer the question.

Hence, the answer is (A).

36. The question is asking for the value of a_4. Substituting $n = 4$ in the formula of the sequence $a_n = (a_{n-1} – 3)^2$ yields $a_4 = (a_{4-1} – 3)^2 = (a_3 – 3)^2$. From this equation, it is clear that we need the value of a_3 to evaluate a_4. Similarly, to evaluate a_3, we need the value of a_2 (a_2 alone is sufficient). Hence, a_4 can be evaluated if we have a_2 alone. Hence, Statement (2) alone is sufficient to answer the question. Similarly, a_2 can also be evaluated from a_1 and therefore the value of a_1 alone is sufficient to evaluate a_4. Since each statement alone is sufficient, the answer is (D).

37. From Statement (1), we have the rule for the general term $a_m = 2m + 1$. Using this formula to evaluate the 5th and 6th terms yields

$$a_5 = 2(5) + 1 = 11$$
$$a_6 = 2(6) + 1 = 13$$

Forming the difference $a_6 – a_5$ yields

$$a_6 – a_5 = 13 – 11 = 2$$

Hence, Statement (1) alone is sufficient to answer the question.

Statement (2) gives the value of the 6th term but does not provide a way to evaluate the 5th term of the series. Hence, Statement (2) alone is not sufficient to answer the question.

Since only Statement (1) alone is sufficient, the answer is (A).

38. From Statement (1), we have the inequality:

$x^2 + 1 > 2x + 4$

$x^2 - 2x + 1 > 4$ subtracting $2x$ from both sides

$x^2 - 2(x)(1) + 1^2 > 4$ expressing the left side in the form $a^2 - 2ab + b^2$

$(x - 1)^2 > 4$ using the formula $a^2 - 2ab + b^2 = (a - b)^2$

Square rooting both sides of the inequality yields two new inequalities: $x - 1 > 2$ or $x - 1 < -2$. Adding 1 to both sides of these inequalities yields $x > 3$ and $x < -1$. So, x is either less than -1 or x is greater than 3. In either case, x does not equal 0. Hence, Statement (1) alone answers the question.

Now, the given inequality in Statement (2) is

$(x + 1)^2 - 2x > 2(x + 1) + 2$

$x^2 + 2x + 1 - 2x > 2x + 2 + 2$

$x^2 + 1 > 2x + 4$

This inequality is same as the inequality in Statement (1). Hence, each statement alone is sufficient to answer the question, and the answer is (D).

39. From statements (1) and (2) together, we have $x = 2y$ and $y = 2k$.

Let k equal -1. Then from the second equation, $y = 2k = 2(-1) = -2$; and from the first equation, $x = 2y = 2(-2) = -4$. In this case, x ($= -4$) is not greater than y ($= -2$).

Now, let k equal 1. Then from the second equation, we have $y = 2k = 2(1) = 2$ and $x = 2y = 2(2) = 4$. In this case, x ($= 4$) is greater than y ($= 2$). We have a double case here with statements (1) and (2) taken together.

Hence, statements (1) and (2) together are not sufficient to answer the question. The answer is (E).

40. The question asks is $y > -1$? We are also given the equation $y = -x$. Substituting this into the inequality $y > -1$ yields $-x > -1$. Multiplying this inequality by -1 and flipping the direction of the inequality yields $x < 1$. Hence, effectively, we are asked "Is $x < 1$."

Now, dividing the inequality in Statement (1) by the positive value x^2 yields $x^2 > 1$. The solutions for this inequality are $x < -1$ and $x > 1$. The first solution says x is less than 1 and the second solution says x is greater than 1. Hence, we have a double case with Statement (1) alone. So, the statement is not sufficient.

Now, dividing the inequality in Statement (2), $x^3 < x^2$, by x^2 yields $x < 1$. Hence, Statement (2) says x is less than 1. Hence, Statement (2) alone is sufficient to answer the question.

The answer is (B).

41. From Statement (1) alone, we have

$3x + y = x + 2y$

$3x + y - x - 2y = 0$ by subtracting x and $2y$ from both sides

$2x - y = 0$

Hence, Statement (1) alone answers the question: $2x - y = 0$.

Now, from Statement (2), we have

$(x + y) + 3(x - y) = 0$

$x + y + 3x - 3y = 0$

$4x - 2y = 0$

$2x - y = 0$

Hence, Statement (2) alone also answers the question: $2x - y = 0$.

Hence, each statement alone is sufficient to answer the question. The answer is (D).

42. Multiplying the equation $1/x + 1/y = 1/3$ given in Statement (1) by xy yields $y + x = xy/3$, or $x + y = xy/3$. This equation is the same as the equation given in Statement (2). Hence, Statement (2) is the same as Statement (1).

Now, multiplying both sides of the equation $x + y = \frac{xy}{3}$ by $\frac{3}{x + y}$ yields $\frac{xy}{x + y} = 3$. Hence, Statement (1) alone, and therefore Statement (2) alone, is sufficient to answer the question.

The answer is (D).

43. $(x - 2)^2 = x^2 - 4x + 2^2 = x^2 - 4x + 4$. Hence, the question is "What is the value of $x^2 - 4x + 4$?"

From Statement (1) alone, we have that $x^2 - 4x + 3 = 0$. Adding 1 to both sides yields $x^2 - 4x + 4 = 1$. The answer is 1. Hence, Statement (1) alone is sufficient.

From Statement (2) alone, we have that $x^2 - 7x + 12 = 0$. Adding $3x - 8$ to both sides yields

$$x^2 - 7x + 12 + (3x - 8) = 0 + 3x - 8, \text{ or } x^2 - 4x + 4 = 3x - 8$$

Here, the value of $x^2 - 4x + 4$ depends on x, an unknown, so Statement (2) alone is not sufficient.

Hence, the answer is (A).

44. 25 percent of x is $25/100 \cdot x$. We can evaluate this if we have the value of x. Hence, the question effectively is "What is the value of x ?"

From Statement (1) alone, we have that 50 percent of x is 50. Hence, we have the equation $\frac{50}{100}x = 50$, or $\frac{x}{2} = 50$, or $x = 100$. Hence, Statement (1) alone is sufficient to answer the question.

From Statement (2) alone, we have that 9/100 of x is 9. Hence, we have the equation $\frac{9}{100}x = 9$, or $x = \frac{100}{9} \times 9 = 100$. Hence, Statement (2) alone is sufficient to answer the question.

Hence, the answer is (D).

45. From Statement (1) alone, we have that x is 50% more than y. Now, 50% of y is $\frac{50}{100}y = \frac{y}{2}$. Hence, $x = y + \frac{y}{2} = \frac{3y}{2}$. Subtracting y from both sides yields $x - y = \frac{y}{2}$. Hence, the value of $x - y$ depends on the value of y, which is unknown. Hence, Statement (1) alone is not sufficient.

From Statement (2) alone, we have that x is the sum of y and 20. Hence, we have the equation $x = y + 20$. Subtracting y from both sides yields $x - y = 20$. Hence, Statement (2) alone is sufficient to answer the question.

Hence, the answer is (B).

46. The two given statements form the following system:

$$x + y = 7$$
$$x^2 + y^2 = 25$$

Solving the top equation for y yields $y = 7 - x$. Substituting this into the bottom equation yields

$$x^2 + (7 - x)^2 = 25$$
$$x^2 + 49 - 14x + x^2 = 25$$
$$2x^2 - 14x + 24 = 0$$
$$x^2 - 7x + 12 = 0$$
$$(x - 3)(x - 4) = 0$$
$$x - 3 = 0 \text{ or } x - 4 = 0$$
$$x = 3 \text{ or } x = 4$$

Now, if $x = 3$, then $y = 7 - 3 = 4$ and $x + y^2 = 3 + 4^2$; and if $x = 4$, then $y = 7 - 4 = 3$ and $x + y^2 = 4 + 3^2$, which does not equal $3 + 4^2$. This is a double case, so the statements together are *not* sufficient. Hence, the statements individually are clearly *not* sufficient. The answer is (E).

47. Multiplying the inequality $x/15 > y/25$ by 15 yields $x > (15/25)y$ or $x > 3y/5$.

Subtracting $2y + 3x$ from both sides of the inequality in Statement (1) yields

$$3y > 5x$$

Rewriting with the variable x on the left and y on the right yields (note that the meaning of the inequality does not change, just the way it is written)

$$5x < 3y$$

Dividing by 5 yields

$$x < 3y/5$$

Hence, Statement (1) alone is sufficient. The answer to the question is "No. x is not greater than $3y/5$."

Now, subtracting $3y + 5x$ from both sides of the inequality in Statement (2) yields

$$3y > 5x$$

Rewriting with the variable x on the left and y on the right yields (note that the meaning of the inequality does not change, just the way it is written)

$$5x < 3y$$

Dividing by 5 yields

$$x < 3y/5$$

Hence, Statement (2) alone is sufficient. The answer to the question is "No. x is not greater than $3y/5$."

Since either statement alone is sufficient, the answer is (D).

48. From Statement (1), we have $\frac{a + b}{2} = 25$.
From Statement (2), we have $\sqrt{ab} = 25$, or by squaring $ab = 25^2$.

Solving the equation $\frac{a + b}{2} = 25$ for b yields $b = 50 - a$. Substituting this into the equation $ab = 25^2$ yields

$$\begin{aligned}
a(50 - a) &= 25^2 \\
a^2 - 50a + 25^2 &= 0 \\
a^2 - 25a - 25a + 25^2 &= 0 \\
a(a - 25) - 25(a - 25) &= 0 \\
(a - 25)(a - 25) &= 0 \\
(a - 25)^2 &= 0 \\
a - 25 &= 0 \\
a &= 25
\end{aligned}$$

Now, $b = 50 - a = 50 - 25 = 25$. Hence, $a = b$ and the statement "$a > b$" is false. Hence, the statements together are sufficient, and the answer is (C).

49. From Statement (1), we have $\frac{a+b}{2} \geq 25$.
From Statement (2), we have $\sqrt{ab} \geq 24$.

The above two equations are symmetric with respect to a and b in the sense that if $a = p$ and $b = q$ is a solution [So, $\frac{p+q}{2} \geq 25$ and $\sqrt{pq} \geq 24$], then $a = q$ and $b = p$ (swapping) is also a solution. Now, in the case $p > q$, the first case ($a = p$ and $b = q$) yields $a > b$ and the second case ($a = q$ and $b = p$) yields $b > a$. Hence, we have a double case, and the statements together are not sufficient. The answer is (E).

50. By the Difference of Squares Formula, we have $x^2 - y^2 = (x - y)(x + y)$.

From Statement (1) alone, we have $x + y = 12$. But we do not have the value of $x - y$, so $x^2 - y^2$ cannot be evaluated.

For example, suppose $x = 6$ and $y = 6$, then $x + y = 12$ and $x^2 - y^2 = 6^2 - 6^2 = 36 - 36 = 0$.

Now, suppose $x = 12$ and $y = 0$. Then $x + y = 12$, and $x^2 - y^2 = 12^2 - 0^2 = 144 - 0 = 144$. Thus, we have a double case, and Statement (1) alone is not sufficient.

From Statement (2) alone, we have the value of $xy = 35$. But again, we do not have the value of $x^2 - y^2$, so $x^2 - y^2$ cannot be evaluated.

For example, suppose $x = 5$ and $y = 7$, then $xy = 35$, and $x^2 - y^2 = 7^2 - 5^2 = 49 - 25 = 24$.

But suppose $x = 35$ and $y = 1$. Then $xy = 35$, and $x^2 - y^2 = 35^2 - 1^2 = 1225 - 1 = 1224$.

Thus, we have a double case and therefore Statement (2) alone is not sufficient.

With the statements together, we have that $x + y = 12$ and $xy = 35$. Squaring both sides of the first equation $x + y = 12$ yields $(x + y)^2 = 12^2 = 144$. Multiplying the second equation $xy = 35$ by 4 yields $4xy = 140$. Subtracting this equation from the equation $(x + y)^2 = 144$ yields

$$\begin{aligned}(x + y)^2 - 4xy &= 144 - 140\\ x^2 + y^2 + 2xy - 4xy &= 4\\ x^2 + y^2 - 2xy &= 4\\ (x - y)^2 &= 4\\ x - y &= \pm 2\end{aligned}$$

Since $x - y$ is either 2 or –2, we still do not have a unique value for $x - y$. Hence, the two statements together are still not sufficient. The answer is (E).

51. $\frac{a+b}{\sqrt{ab}} \geq \frac{a}{\sqrt{ab}} + \frac{b}{\sqrt{ab}} \geq \frac{\sqrt{a^2}}{\sqrt{ab}} + \frac{\sqrt{b^2}}{\sqrt{ab}} \geq \sqrt{\frac{a^2}{ab}} + \sqrt{\frac{b^2}{ab}} \geq \sqrt{\frac{a}{b}} + \sqrt{\frac{b}{a}} \geq \sqrt{\frac{a}{b}} + \frac{1}{\sqrt{a/b}}$. Hence, we need only the value of a/b to evaluate the expression.

From Statement (1) alone, we have that a is 4 times b. Hence, $a = 4b$, or $a/b = 4$. Since we can derive the value of a/b from Statement (1) alone, the statement is sufficient.

From Statement (2) alone, we have that b is one-fourth of a. Hence, we have $b = a/4$. Multiplying both sides by $4/b$ yields $4 = a/b$. Since we can derive the value of a/b from Statement (2) alone, the statement is sufficient.

The answer is (D).

52. Adding the equation $yz - zx = 3$ from Statement (1) and the equation $zx - xy = 4$ from Statement (2) yields

$$(yz - zx) + (zx - xy) = 3 + 4$$
$$yz - zx + zx - xy = 7$$
$$yz - xy = 7$$
$$xy - yz = -7 \qquad \text{multiplying both sides by } -1$$

Hence, the statements together are sufficient to answer the question.

The answer is (C).

53. Statement (1) alone, says that $x^5 > y^5$. In the case of odd exponents (here 5), we can drop the exponents from both sides and conclude that $x > y$. Hence, Statement (1) alone is sufficient.

Statement (2) alone, says that $x^4 < y^4$. We cannot drop even exponents from both sides. Hence, Statement (2) alone is not sufficient.

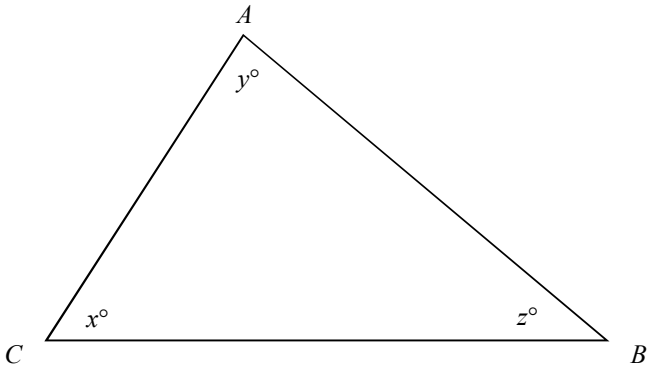
For example, suppose $x = 1$ and $y = 2$. Here, $x < y$ and $x^4 < y^4$.

Now, Suppose $x = 1$ and $y = -2$. Here, $x > y$ and $x^4 < y^4$.

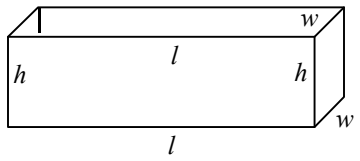
We have a double case. Hence, Statement (2) alone is not sufficient.

The answer is (A).

54. Since with the statements together we have only two of the three angles and no side length, statements (1) and (2) together are not sufficient. The answer is (E). For example, the triangle shown below satisfies statements (1) and (2), but has a greater length for side *AB* than in the original figure.



55. Let the rectangular box be as shown in the figure:



The surface area of the box is $lw + 2lh + 2hw$, and the volume is lwh.

From Statement (1), we have that the area of the base, which equals lw, is 23.

From Statement (2), we have that the volume of the box, which equals lwh, is 23.

Dividing the value in Statement (2) by the value in Statement (1) yields

$$\frac{lwh}{lw} = h = \frac{23}{23} = 1$$

So, we have the height of the box. Since we have the values of only h and lw, there is not enough information to determine the value of the expression $lw + 2lh + 2hw$, which has to be evaluated.

Hence, the statements together are insufficient to answer the question.

The answer is (E).

56. Factoring the common factor b from the expression $ab + bc$ yields $b(a + c)$.

From Statement (1) alone, we have $b = a + c$. Replacing $a + c$ with b in the expression $b(a + c)$ yields $b \cdot b$, or b^2. Since we do not have the value of b or $a + c$, which equals b, the expression cannot be evaluated. Hence, Statement (1) alone is not sufficient.

From Statement (2) alone, we have that $b = 3$. Substituting this in the expression $b(a + c)$ yields the expression $3(a + c)$. Since we do not know the value of $a + c$, we cannot evaluate the expression. Hence, Statement (2) alone is not sufficient.

Now, let's combine the statements. From Statement (1), we have that the expression $b(a + c)$ equals b^2; and since from Statement (2) we have that b equals 3, the value of the expression is $b^2 = 3^2 = 9$. Hence, the answer is (C).

57. We are given that $(x - y)(x + y) = 15$ …(A)

From Statement (1), we have $x - y = 3$. Squaring both sides yields $(x - y)^2 = 3^2 = 9$. Dividing equation (A) by this equation yields $\frac{(x-y)(x+y)}{(x-y)^2} = \frac{15}{9}$, or $\frac{x+y}{x-y} = \frac{15}{9}$.

Since we are looking for the ratio x/y, let's divide every variable on the left-hand side of the equation by y, thereby creating the ratio x/y:

$$\frac{\frac{x}{y}+\frac{y}{y}}{\frac{x}{y}-\frac{y}{y}} = \frac{15}{9}$$

$$\frac{\frac{x}{y}+1}{\frac{x}{y}-1} = \frac{15}{9}$$

$$9\left(\frac{x}{y}+1\right) = 15\left(\frac{x}{y}-1\right)$$

$$9\frac{x}{y}+9 = 15\frac{x}{y}-15$$

$$6\frac{x}{y} = 24$$

$$\frac{x}{y} = 4$$

Hence, Statement (1) alone is sufficient.

From Statement (2), we have $x + y = 5$. Squaring both sides yields $(x + y)^2 = 5^2 = 25$. Dividing equation (A) by this equation yields $\frac{(x-y)(x+y)}{(x+y)^2} = \frac{15}{25}$, or $\frac{x-y}{x+y} = \frac{15}{25}$. Solving the equation for x/y as above yields $x/y = 4$. Hence, Statement (2) alone is sufficient.

Since either statement is sufficient, the answer is (D).

58. Compound interest is the concept of adding accumulated interest to the principal so that interest is earned on the new principal from that moment onward. The act of declaring interest to be principal is called compounding (i.e., interest is compounded).

To calculate the interest earned or the balance at any point, apart from the interest earned, we also need the frequency of compounding of the interest. Hence, both statements are required to answer the question. The answer is (C).

59. Statement (1) is just one constrain in 2 unknowns, and Statement (2) is also just one constrain in 2 unknowns. To solve for the two unknowns m and n we need at least 2 constraints. So, either statement alone is not sufficient.

Using both statements, we have $7m + 5n = 29$ and m and n are positive integers. Suppose $p = 7m$ and $q = 5n$. Then $p + q = 29$ and $q = 29 – p$ [(a positive multiple of 5) equals 29 – (a positive multiple of 7)]. Since m and n are positive integers, p is a multiple of 7 and q is a multiple of 5. Now,

If $p = 7$, $q = 29 – 7 = 22$, not a multiple of 5. Reject.

If $p = 14$, $q = 29 – 14 = 15$, a multiple of 5. Accept.

If $p = 21$, $q = 29 – 21 = 8$, not a multiple of 5. Reject.

If $p = 28$, $q = 29 – 28 = 1$, not a multiple of 5. Reject.

If $p \geq 35$, $q \leq 29 – 35 = –6$, not positive. Reject.

So, we have a unique solution $m = p/7 = 14/7 = 2$ and $n = q/5 = 15/5 = 3$. Hence, the answer is (C).

60. Two lines (here l and m) cut by a third line (a transversal, here k) are parallel when any one of the following can be shown:

(1) Corresponding angles are equal: $y = q$. Hence, Statement (2) alone is sufficient.
(2) Alternate interior angles are equal: No information is given about such angles in the problem.
(3) Alternate exterior angles are equal: No information is given about such angles in the problem.
(4) Interior angles are supplementary. $y + p = 180°$. No information is given about such the angles in problem.
(5) Exterior angles are supplementary. $x + q = 180°$. We are told that the exterior angles x and q are equal in Statement (1), but this does not determine whether their sum is 180˚.

Hence, Statement (1) alone is *not* sufficient, and Statement (2) alone is sufficient. The answer is (B).

61. $x/4$ is a positive integer when x is a multiple of 4. Now, not every multiple of 4 is divisible by 14. For example, 28 is a multiple of 4 and divisible by 14, but 32 is a multiple of 4 and not divisible by 14. Hence, Statement (1) alone is not sufficient.

$x/6$ is a positive integer when x is a multiple of 6. Now, not every multiple of 6 is divisible by 14. For example, 42 is a multiple of 6 and divisible by 14, but 48 is a multiple of 6 and not divisible by 14. Hence, Statement (2) alone is not sufficient.

With statements (1) and (2) together, we have that x is a multiple of both 4 and 6. Hence, x must be a multiple of the least common multiple of 4 and 6, which is 12. Hence, x is a multiple of 12. Now, not every multiple of 12 is divisible by 14. For example, 84 is a multiple of 12 and divisible by 14, but 24 is a multiple of 12 and not divisible by 14. Hence, even statements (1) and (2) together do not answer the question. The answer is (E).

62. Remember that *Average Speed = Net Distance ÷ Time Taken.* We are given that the time taken for the full trip is 30 minutes. Hence, we need only the distance traveled.

From Statement (1) alone, we have that the restaurant is 2 miles from home. Since Waugh jogs back along the same route, the net distance he traveled is 2 miles + 2 miles = 4 miles. Hence, his average speed is 4 miles ÷ 30 minutes, which can be calculated. Hence, Statement (1) alone is sufficient.

From Statement (2) alone, we have that $x = 12$ and $y = 6$. Let d be the distance to the restaurant. Then the time taken to jog to the restaurant, by the formula *Time = Distance ÷ Speed*, equals $d \div x = d/12$; and to return home from restaurant, the time taken equals $d \div y = d \div 6$. Hence, the total time taken equals $d/12 + d/6$, which is given to be 30 minutes (1/2 hour). Hence, d can be calculated from this formula, and therefore the average speed can be calculated using *Distance ÷ Time* = $\frac{2d}{\frac{1}{2}\text{ hours}} = 4d$. Hence, Statement (2) alone is also sufficient.

Hence, the answer is (D).

63. The shaded region contains two triangles ΔAED and ΔEBC. Hence, the area of the shaded region equals the sum of the areas of these two triangles.

Now, the formula for the area of a triangle is $1/2 \times base \times height$. Hence, the area of $\Delta AED = 1/2 \times AE \times AD$, and the area of $\Delta EBC = 1/2 \times EB \times BC = 1/2 \times EB \times AD$ (since $BC = AD$).

So, the area of the shaded region equals $1/2 \times AE \times AD + 1/2 \times EB \times AD = 1/2 \times AD(AE + EB) = 1/2 \times AD \times AB$ (since $AB = AE + EB$). Hence, to evaluate the area of the shaded region, we need AD and AB, which are given in Statement (1). Hence, Statement (1) alone is sufficient.

Statement (2) gives the lengths of AE and EB, which are only horizontal lengths. Since we do not have any vertical lengths (AD and BC) in the figure, the vertical lengths are free to take any values and therefore Statement (2) alone is *not* sufficient.

Hence, the answer is (A).

64. The formula for the area of an equilateral triangle of side length a is $\frac{\sqrt{3}}{4}a^2$. Since the only unknown in the equation is the side length a, given the value, the area can be determined. Hence, Statement (1) alone, which gives the side length BC, is sufficient.

From Statement (2) alone, we have the length of an altitude of the triangle. The formula for the side length of an altitude of an equilateral triangle of side length a is $\frac{\sqrt{3}}{2}a$. So, if d is the length of the altitude, then $d = \frac{\sqrt{3}}{2}a$. Solving the equation for a yields $a = \frac{2d}{\sqrt{3}}$. Hence, even if we have just an altitude, the side length, and therefore the area of the triangle, can be evaluated. Hence, Statement (2) alone, which gives the altitude length AD, is sufficient.

Hence, the answer is (D), either statement is sufficient.

65. We are given the expression $x^2 - y^2 = 15$. Applying the Difference of Squares formula $(a + b)(a - b) = a^2 - b^2$ to the left-hand side yields $(x + y)(x - y) = 15$.

From Statement (1) alone, we have $x - y = 5$. Plugging this in the equation $(x + y)(x - y) = 15$ yields $(x + y)(5) = 15$. Dividing both sides by 5 yields $x + y = 3$. Hence, we have the values of both $x + y$ and $x - y$. Now, we can compare the two and answer the question. So, Statement (1) alone is sufficient. The answer is "No. $x + y > x - y$ is false."

From Statement (2) alone, we have $x + y = 3$. Plugging this in the equation $(x + y)(x - y) = 15$ yields $3(x - y) = 15$. Dividing both sides by 3 yields $x - y = 5$. Hence, we have the values of both $x + y$ and $x - y$. Now, we can compare the two and answer the question. Hence, Statement (2) alone is sufficient. The answer to the question is "No. $x + y$ is not greater than $x - y$."

Hence, the answer is (D).

66. From Statement (1) alone, we have the inequality $x > 2$.

Now, suppose $x = 2.5$. Then $x - 2 = 2.5 - 2 = 0.5$, and $x - 3 = 2.5 - 3 = -0.5$. Hence, $(x - 2)(x - 3) = (0.5)(-0.5) = -0.25$. So, $(x - 2)(x - 3)$ is negative.

Now, suppose $x = 4$. Then $x - 2 = 4 - 2 = 2$, and $x - 3 = 4 - 3 = 1$. Hence, $(x - 2)(x - 3) = 2 \cdot 1 = 2$. So, $(x - 2)(x - 3)$ is positive.

Hence, we have a double case, and Statement (1) alone is not sufficient.

From Statement (2) alone, we have the inequality $x < 3$.

Now, suppose $x = 2.5$. Then $x - 2 = 2.5 - 2 = 0.5$ and $x - 3 = 2.5 - 3 = -0.5$. Hence, $(x - 2)(x - 3) = (0.5)(-0.5) = -0.25$. So, $(x - 2)(x - 3)$ is negative.

Now, suppose $x = 1$. Then $x - 2 = 1 - 2 = -1$ and $x - 3 = 1 - 3 = -2$. Hence, $(x - 2)(x - 3) = (-1)(-2) = 2$. So, $(x - 2)(x - 3)$ is positive.

Hence, we have a double case, and Statement (2) alone is not sufficient.

Now, with statements (1) and (2) together, we have $x > 2$ and $x < 3$. Subtracting 2 from both sides of first inequality and 3 from both sides of the second inequality yields $x - 2 > 0$ and $x - 3 < 0$. Hence, $x - 2$ is positive, and $x - 3$ is negative. Since the product of a positive and a negative number is always negative, we have $(x - 2)(x - 3)$ is negative and therefore less than 0. Hence, $(x - 2)(x - 3)$ is not greater than 0, and the answer to the question is "No. $(x - 2)(x - 3)$ is less than 0."

Since both statements were required, the answer is (C).

67. If you divided the inequality $x > y$ by xy (positive with the two statements together) and derived that $1/x > 1/y$ and therefore answered (C), you are mistaken. There is an deeper study to make.

From Statement (1) alone, we have x is negative. Since $x > y$, y must also be negative. Hence, xy, the product of two negative numbers, must be positive. Dividing the inequality by the positive expression xy yields $x/xy > y/xy$, or $1/y > 1/x$. Rearranging yields $1/x < 1/y$. Hence, Statement (1) alone is sufficient. The answer to the question is "Yes, $1/x < 1/y$."

From Statement (2) alone, we have y is negative. The only restriction on x is $x > y$ (given), so x may be negative, zero, or positive.

In case x is positive and since y is negative, xy is negative. Dividing the given inequality $x > y$ by xy (a negative value) and flipping the direction of inequality yields $1/y < 1/x$.

In case both x and y are negative, xy is positive and here we can divide the inequality $x > y$ by xy without flipping the direction of inequality. Thus, here we get $1/y > 1/x$.

Hence, we have a double case with Statement (2) alone.

Since Statement (1) alone is sufficient and Statement (2) alone is *not*, the answer is (A).

68. Let l be the length of the rectangle. Then $l = 3.2$ yards. The formula for the area of a rectangle is *length* × *width*. Hence, the area of the rectangle is $lw = 3.2w$. So, to evaluate the area, we need only the value of w. Hence, the question can be modified as "Width of rectangle = ?"

From Statement (1) alone, we have the length of fence required (perimeter) for the field is 11.2 yards. The formula for the perimeter of a rectangle is 2(*length* + *width*). Hence, the perimeter of the field is $2(l + w) = 11.2$, or $2(3.2 + w) = 11.2$. Solving for w yields $w = 5.6 - 3.2 = 2.4$. Hence, Statement (1) alone is sufficient.

From Statement (2) alone, we have that the distance between the diagonally opposite corners (diagonal length) of the rectangle field is 4 yards. A diagonal of a rectangle forms a hypotenuse with the length and width of the rectangle. Hence, by The Pythagorean Theorem, $l^2 + w^2 = (\text{diagonal length})^2 = 4^2$, or $3.2^2 + w^2 = 4^2$. Solving this equation for w yields $w = \sqrt{4^2 - 3.2^2} \approx 2.4$. Hence, Statement (2) alone is also sufficient.

The answer is (D).

69. Since OB is a radius, B is a point on the circle. Now, BC is a tangent to the circle only if $\angle CBO$ is right angled. In this case, ΔCBO is right triangle, and by The Pythagorean Theorem $OC^2 = OB^2 + BC^2$. Hence, the question can be reduced to "Is $OC^2 = OB^2 + BC^2$?"

We know that $OB = 3$ (given in figure). Now, we need the other two sides OC and BC to answer the question.

From Statement (1), we have $PC = 2$, and we know $OP = radius = 3$. Hence, $OC = OP + PC = 3 + 2 = 5$. But, we still do not have the value of BC. Hence, Statement (1) is *not* sufficient.

From Statement (2), we have the area of the square is 16. By the formula, the area of a square = $side^2$, we have $BC^2 = 16$; and by square rooting, we have $BC = 4$. But, from this Statement alone, we cannot determine the value of OC. Hence, Statement (2) alone is *not* sufficient.

From the statements together, we have $OC = 5$, $BC = 4$, and we already have $OB = 3$. Now, we have all the details to determine whether OC^2 equals $OB^2 + BC^2$. Hence, both statements are required, and the answer is (C).

70. An integer is called prime if the only positive integers that divide it evenly are 1 and the number itself.

From Statement (1), we have that x and y are two prime numbers. So, xy does not equal x or y. Now, xy divided by x equals y (an integer), and xy divided by y equals x (an integer). Hence, xy has two more factors other than 1 and xy itself. So, by the definition of a prime number, xy is *not* prime. Hence, Statement (1) alone is sufficient to answer the question.

From Statement (2), we have that x and y are two odd numbers not equal to 1. So, xy does not equal x or y. Now, xy divided by x equals y (an integer), and xy divided by y equals x (an integer). Hence, xy has at least two more factors other than 1 and xy itself. So, by the definition of a prime number, xy is *not* prime. Hence, Statement (2) alone is sufficient to answer the question.

The answer is (D).

71. From Statement (1) alone, we have the inequality $x < 2$. So, $x - 2$ is negative. So, $x - 3 = x - 2 - 1$ = a Negative Number – 1 = a Negative Number. So, $(x - 2)(x - 3)$ = a Negative Number × a Negative Number = a Positive Number. Hence, Statement (1) alone is sufficient.

From Statement (2) alone, we have the inequality $x < 3$.

Now, suppose $x = 2.5$. Then $x - 2 = 2.5 - 2 = 0.5$ and $x - 3$ equals $2.5 - 3 = -0.5$. Hence, $(x - 2)(x - 3) = (0.5)(-0.5) = -0.25$. So, $(x - 2)(x - 3)$ is negative.

Now, suppose $x = 1$. Then $x - 2 = 1 - 2 = -1$, and $x - 3 = 1 - 3 = -2$. Hence, $(x - 2)(x - 3) = (-1)(-2) = 2$. So, $(x - 2)(x - 3)$ is positive.

Hence, we have a double case, and Statement (2) alone is *not* sufficient.

The answer is (A), Statement (1) alone sufficient and Statement (2) alone *not* sufficient.

72. To calculate an average, we need exact quantities such as the distribution of the numbers, the averages across sections, etc. But from statements (1) and (2) together, we have only the ranges: Half of the test takers scored above 500, and the lowest and the highest scores on the test are 200 and 800. These are not exact quantities and therefore do not help us evaluate an exact quantity such as an average. Hence, the statements together are not sufficient. The answer is (E).

73. Dividing the equation $6p + 4q = 10$ in Statement (2) by 2 yields $3p + 2q = 5$; this is the same equation in Statement (1). Hence, Statement (1) here is the same as Statement (2) put in another form. Now, if a solution is possible, it should be possible by either statement alone. Otherwise, it is not possible with the statements together. Hence, the possible answers are (D) and (E).

Now, multiplying the equation $3p + 2q = 5$ by 2/3 yields $2p + 4q/3 = 10/3$. Subtracting $q/3$ from both sides yields $2p + q = 10/3 - q/3$. Since q is unknown, $2p + q$ cannot be evaluated. Hence, the statements together are not sufficient. The answer is (E).

74. From Statement (1) alone, we have that x is 1 unit greater than y. Hence, Statement (1) alone answers the question. The answer is "Yes. x is greater than y."

From Statement (2) alone, we have the equation $x + y = 1$. This equation is symmetric in x and y. Hence, the relation between x and y cannot be established. For example,

If $x = 1$ and $y = 0$, then $x + y = 1 + 0 = 1$ and $x > y$.

If $x = 0$ and $y = 1$, then $x + y = 0 + 1 = 1$ and $x < y$.

Hence, we have a double case, and Statement (2) alone is not sufficient to answer the question.

Hence, the answer is (A).

75. We are given that $x = y/7$.

From Statement (1) alone, we have $y > 210$. Dividing this inequality by 7 yields $y/7 > 210/7 = 30$. Replacing $y/7$ with x yields $x > 30$. Hence, x is not less than 30. Hence, Statement (1) alone is sufficient to answer the question.

From Statement (2) alone, we have $y < 217$. Dividing this inequality by 7 yields $y/7 < 217/7 = 31$. Replacing $y/7$ with x yields $x < 31$. Now, suppose x equals 29 ($x < 31$), then x is less than 30. Now, suppose $x = 30$ ($x < 31$), then x is not less than 30. Hence, we have a double case, and Statement (2) alone is not sufficient to answer the question.

Hence, the answer is (A).

76. The question is about the fraction $\frac{|a-b|+|c-b|}{d}$. The numerator of the fraction is $(a - b) + (c - b)$ and the denominator is d.

From Statement (1), we do not have any information about d. So, if $(a - b) + (c - b)$ is negative and d is positive, the fraction is negative; and if $(a - b) + (c - b)$ is negative and d is negative, the fraction is positive. Hence, Statement (1) alone is *not* sufficient.

From Statement (2), we do not have information about the numerator $(a - b) + (c - b)$. So, if the numerator is negative and the denominator d is negative, the fraction is positive; and if the numerator is positive and the denominator d is negative, the fraction is negative. Hence, we have a double case and Statement (2) alone is *not* sufficient.

Now, the numerator is $(a - b) + (c - b) = a + c - 2b$ and according to Statement (1) alone, we have the expression is negative (< 0). Now, from statement (2) alone, we have the denominator d is negative. So, the fraction Numerator/Denominator = a Negative Number ÷ a Negative Number = a Positive Number. So, the answer is "Yes. $\frac{|a-b|+|c-b|}{d} \geq 0$". Hence, the answer is (C), the statements together are sufficient.

77. From Statement (1) alone, we have that $m/14$ is an integer. Now, suppose $m = 42$. Then $m/14 = 3$, a positive integer.

Now, $m = 42 = 2 \cdot 3 \cdot 7$. The prime factors of m are 2, 3, and 7. Hence, the number of prime factors of m is 3. Here, the answer is "No. m does not have even 4 prime factors."

Now, suppose $m = 210 = 2 \cdot 3 \cdot 5 \cdot 7$. Here, $m/14 = 3 \cdot 5$, a positive integer, and m has 4 prime factors: 2, 3, 5, and 7.

Here, the answer is "Yes. m has exactly 4 prime factors." Hence, we have a double case, and Statement (1) alone is not sufficient.

From Statement (2) alone, we have that $m/15$ is an integer.

Now, suppose $m = 30$. Then $m/15 = 2$, a positive integer. Now, $m = 30 = 2 \cdot 3 \cdot 5$, so the prime factors of m are 2, 3, and 5. Hence, the number of prime factors of m is 3. Here, the answer is "No. m does not have even 4 prime factors."

Now, suppose $m = 2 \cdot 3 \cdot 5 \cdot 7 = 210$. Here, $m/15 = 2 \cdot 7$, a positive integer, and m has 4 prime factors: 2, 3, 5, and 7. Here, the answer is "Yes. m has exactly 4 prime factors." Hence, we have a double case, and Statement (2) alone is not sufficient.

Now, with statements (1) and (2) together, we have that m is divisible by both 14 and 15. Hence, m is a multiple of both 14 and 15. Hence, m must equal the least common multiple of 14 and 15 or its multiple. Let's assume that m is only equal to the LCM. The LCM of 14 (= 2 · 7) and 15 (= 3 · 5) is 2 · 7 · 3 · 5 (= 210). The LCM itself has 4 prime factors 2, 7, 3, and 5. Hence, its multiples might have more prime factors. Hence, both statements are required to answer that m has at least 4 prime factors. Hence, the statements together answer the question. The answer is (C).

78. From Statement (1) alone, we have that $n/10$ is an integer. Hence, n is either equal to 10 or a multiple of 10.

Suppose $n = 10$. The number 10 factors into the primes 2 · 5. Hence, n has exactly 2 (not at least 3) prime factors. Here, n does not even have 3 prime factors.

Now, suppose $n = 30$. Then $n/10$ equals 3 an integer. Now, n factors into the primes 2 · 3 · 5 and has exactly 3 prime factors.

Hence, we have a double case, and Statement (1) alone is not sufficient to answer the question.

From Statement (2) alone, we have that $n/30$ is an integer. Hence, n is equal to either 30 or a multiple of 30. The number 30 factors into the primes 2 · 3 · 5, and multiples of 30 may have even more prime factors. Hence, n has at least 3 prime factors 2, 3, and 5. So, Statement (2) alone is sufficient to answer the question.

The answer is (B).

79. From Statement (1) alone, we have that $y = \frac{1}{x-1}$. For any value of x, $\frac{1}{x-1}$ cannot equal 0. For example, suppose $\frac{1}{x-1} = 0$. Then multiplying the equation by $x - 1$ yields a false result 1 = 0. Hence, the assumption that $y = \frac{1}{x-1} = 0$ is false. Hence, Statement (1) alone is sufficient to answer the question. The answer is "No. y is not equal to 0."

From Statement (2) alone, we have that $y = x/6$. Now, suppose $x = 0$. Then $y = x/6 = 0/6 = 0$. But if x is not zero, say $x = 2$, then $y = x/6 = 2/6 \neq 0$. Hence, we have a double case, and therefore Statement (2) alone is not sufficient to answer the question.

The answer is (A).

80. Savings (in the current context) equals Income – Expenditure.

Statement (1) has information only about income, and Statement (2) has information only about expenditure. Hence, neither statement alone is sufficient to answer the question.

From the statements together, suppose Kelvin's income is \$9 in 1998 and \$12 in 1999 [the incomes are in accordance with the ratio in Statement (1)]. Also suppose his expenditure is \$5 in 1998 and \$6 in 1999 [the expenditures are in accordance with the ratio in Statement (2)]. Then the ratio of his savings in the year 1998 to 1999 would be (\$9 – \$5) : (\$12 – \$6) = \$4 : \$6 = 2 : 3.

In another case, suppose Kelvin's income is \$12 in 1998 and \$16 in 1999 [the incomes are in accordance with the ratio in Statement (1)]. Also suppose his expenditure is \$10 in 1998 and \$12 in 1999 [the expenditures are in accordance with the ratio in Statement (2)]. Then the ratio of his savings in 1998 to 1999 would be (\$12 – \$10) : (\$16 – \$12) = \$2 : \$4 = 1 : 2. We have a different savings ratio here. Hence, we already have a double case and therefore the two statements together do not determine a unique ratio for savings.

Hence, the answer is (E).

81. From Statement (1), we have information about the average temperature only for the period Monday through Wednesday. Since there is no relevant information for the rest of the week, Statement (1) alone is not sufficient.

Statement (2) has information about the minimum and the maximum temperatures for the period Thursday through Saturday. It does not help us evaluate the average temperature for the period.

With two statements together, we do not have information about average temperatures for the complete week. Hence, the statements together are not sufficient to answer the question.

The answer is (E).

82. We are given the inequality $a < b$.

From statement (1) alone, we have $c < a$. Combining this with the inequality $a < b$ yields $c < a < b$. Picking the first and the last parts we have the inequality $c < b$ [Same as Statement (2)]. We will use this inequality to see whether the given expression is positive. Adding the inequality to the known inequality $a < b$ yields $a + c < 2b$. Subtracting $2b$ from both sides yields $a + c - 2b < 0$. Hence, $(a - b) + (c - b) = a + c - 2b$ is less than 0 and is *not* positive. Hence, Statement (2) alone, derived from Statement (1), is also sufficient. The answer is (D).

➢ **Hard**

83. Statement (1) contains two variables, but only one equation. Hence, we cannot determine the values of p and q using the statement alone.

Similarly, we cannot determine the value of the given expression from Statement (2) alone.

Together, statements (1) and (2) form a system of two equations in two unknowns that may have a solution. Solving the equation $pq = 35$ for q yields $q = 35/p$. Plugging this into the equation $p + q = 12$ yields

$p + 35/p = 12$

$p^2 + 35 = 12p$ by multiplying both sides by p

$p^2 - 12p + 35 = 0$ by subtracting $12p$ from both sides

$(p - 5)(p - 7) = 0$

$p - 5 = 0$ or $p - 7 = 0$

$p = 5$ or $p = 7$

When $p = 5$, the equation $p + q = 12$ yields $q = 7$. Similarly, when $p = 7$, $q = 5$. Hence, the solutions of the system are the ordered pairs (5, 7) and (7, 5). From the symmetry in the expression $\frac{1}{p^4} \square \frac{1}{q^4}$, both ordered pairs will return the same value when plugged into the expression. Thus, the value of the expression $\frac{1}{p^4} \square \frac{1}{q^4}$ can be determined only by using both statements. The answer is (C).

84. The expression on the right-hand side of the equation cannot be further reduced. Hence, the variable H (the only variable on the left-hand side) is dependent only on the variables: T, R, k, and g (the only variables on the right-hand side). We can collect the values for all four variables on the right-hand side only by using both statements and thereby determine the value of H. Hence, both statements are required, and the answer is (C).

85. We are asked to determine whether the equation $\sqrt{H^2 - 2H} \stackrel{?}{=} \frac{T\sqrt{1-R^2}}{k\sqrt{1-g^k}}$ is true.

We are given that $H = 3$. Hence, we can evaluate the left-hand side.

There are four unknowns in the right-hand side: T, R, k, and g. If we know the values of these unknowns, then the right-hand side can also be evaluated and we can determine whether the equation is true. We need both statements to know the values of the four unknowns. Hence, the statements together are sufficient to answer the problem. The answer is (C).

86. Let the number of girls be g, and the number of boys be b. Hence, the number of students is $b + g$. From the statements together, we have that

10% of girls are blue eyed. 10% of g is $\frac{10}{100}g \stackrel{?}{=} \frac{g}{10}$.
20% of boys are blue eyed. 20% of b is $\frac{20}{100}b \stackrel{?}{=} \frac{b}{5}$.

Hence, the total number of blue-eyed students is $\frac{g}{10} \stackrel{?}{=} \frac{b}{5}$.

Hence, the required fraction is $\frac{\frac{g}{10} \stackrel{?}{=} \frac{b}{5}}{b \stackrel{?}{=} g}$.

Dividing both the numerator and the denominator of the fraction by b yields $\frac{\frac{g}{10b} \stackrel{?}{=} \frac{1}{5}}{1 \stackrel{?}{=} \frac{g}{b}}$. Hence, to find the value of the required fraction, we need at least the ratio of b/g. Since we do not have means to evaluate the ratio, the statements together are not sufficient to answer the question. The answer is (E).

87. If n is odd, then both $n + 1$ and $n + 3$ are even and their product $(n + 1)(n + 3)$ is a multiple of 4 (For example, $6 \times 10 = 60$, a multiple of 4).

If n is even, then both $n + 1$ and $n + 3$ are odd and their product $(n + 1)(n + 3)$ is odd, which is not a multiple of 4.

These are the only two significant cases.

Hence, $(n + 1)(n + 3)$ is a multiple of 4 only when n is odd. Hence, the question can be reduced to "Is n odd?"

Now, from Statement (1) alone, we have that $(n + 2)(n + 4)$ is odd. Hence, we have

$(n + 2)(n + 4) = \text{Odd}$

$n^2 + 6n + 8 = \text{Odd}$

$n^2 + 2(3n + 4) = \text{Odd}$

$n^2 + \text{Even Number} = \text{Odd}$

$n^2 = \text{An Odd Number} - \text{An Even Number} = \text{An Odd Number}$

$n^2 = \text{Square Root of an Odd Number} = \text{Odd Number}$

Hence, Statement (1) alone is sufficient to answer the question: "Is n odd?". The answer is "Yes."

From Statement (2) alone, we have that $(n + 3)(n + 6)$ is even.

Now, suppose n is even. Then $n + 3$ must be odd and $n + 6$ must be even and the product $(n + 3)(n + 6)$ must be even. Now, suppose n is odd. Then $n + 3$ must be even and $n + 6$ must be odd and the product $(n + 3)(n + 6)$ is even, again.

Hence, $(n + 3)(n + 6)$ is even, whether n is odd or whether n is even. Hence, Statement (2) alone is *not* sufficient to answer the question.

The answer is (A).

88. From Statement (1) alone, we have that all 4-digits of X are greater than 5. Hence, each digit must be at least 6 or greater. Hence, the sum of its digits must be at least $4 \times 6 = 24$.

Now, suppose $Y = 111$. Then the sum of its digits is $1 + 1 + 1 = 3$, which is less than 24 (a possible value of the sum of the digits of X).

Now, suppose $Y = 999$. Then the sum of its digits is $9 + 9 + 9 = 27$, which is greater than 24 (a possible value of the sum of the digits of X).

Hence, we have a double case, and Statement (1) alone is *not* sufficient.

From Statement (2) alone, we have that the 3 digits of Y are less than 5. Hence, none of the digits is greater than 4. Hence, the sum of the digits of Y is not greater than $3 \times 4 = 12$.

Now, suppose $X = 1111$. Then the sum of its digits is $1 + 1 + 1 + 1 = 4$, which is less than 12 (a possible value of the sum of the digits of Y).

Now, suppose $X = 4444$. Then the sum of its digits is $4 + 4 + 4 + 4 = 16$, which is greater than 12 (a possible value of the sum of the digits of Y).

Hence, we have a double case, and Statement (2) alone is *not* sufficient.

From the statements together, we have that the sum of the digits of X is greater than or equal to 24, and the sum of the digits of Y is less than or equal to 12. Hence, the sum of the digits of X is greater than the sum of the digits of Y. Hence, the statements together answer the question, and the answer is (C).

89. Let E represent the total number of employees. Let w represent the number of workers, x the number of executives, and c the number of clerks. Then we have

$$E = w + x + c \qquad (1)$$

From Statement (1) alone, we have that 41% of the employees E are workers (w), 460 are executives (x), and the remaining 720 employees are clerks (c). Hence, we have the equations

$$w = 41/100 \times E \qquad (2)$$
$$x = 460 \qquad (3)$$
$$c = 720 \qquad (4)$$

With equations (1), (2), (3), and (4), we have four different linear equations in four unknowns E, w, x, and c. Hence, a unique solution must be possible and can be determined. Hence, Statement (1) alone is sufficient. The answer to the question is "$E = 2000$."

From Statement (2) alone, we have that the number of executives (x) is 460, and this is 23% of the total number of employees (E). Hence, we have

$$23\% \text{ of } E = x = 460$$

$$(23/100)E = 460$$

$$E = 460/23 \times 100 = 2000$$

Hence, Statement (2) alone is sufficient.

Since either statement (1) or (2) is sufficient, the answer is (D).

90. Let A be the set of families having exactly one car. Then the question is how many families are there in set A.

Next, let B be the set of families having exactly two cars, and let C be the set of families having more than two cars.

Then the set of families having at least one car is the collection of the three sets A, B, and C. Since Statement (1) gives only the total number of families in the three sets, the statement is not sufficient.

Next, the set of families having at least two cars is the collection of the two sets B and C. Since Statement (2) does not have any information about set A, Statement (2) alone is not sufficient.

Now, from the statements together we have that

The number of families in all the three sets A, B, and C together = 250 [from Statement (1)]

and

The number of families in the two sets B and C together = 60 [from Statement (2)]

Since set A is the difference between a set containing the three families of A, B, and C and a set of families of B and C only, The number of families in set A =

The number of families in sets A, B, and C together – the number of families in sets B and C =

$$250 - 60 =$$

$$190$$

Hence, both statements are required to answer the question, and the answer is (C).

91. Let p be the population of Paxton in 1960.

Then from Statement (1) alone, the population increased by 10,082 in the first decade. So, the new population is $p + 10{,}082$. Hence, the percentage increase in the population is $10{,}082/p \times 100$. Since we do not know the population increase in the second decade, we cannot calculate the percentage increase and therefore Statement (1) alone is not sufficient.

From Statement (2) alone, we have that the population increase was the same (say, i) in each decade. Hence, the population should be $p + i$ at the end of the year 1969 and $(p + i) + i = p + 2i$ at the end of the second decade. So, the percentage increase in population in the first decade equals

$$\frac{\text{Increase in population in the first decade}}{\text{The population in 1960}} \times 100 = \frac{i}{p} \times 100$$

Also, the percentage increase in population in the second decade equals

$$\frac{\text{Increase in population in the second decade}}{\text{The population in 1970}} \times 100 = \frac{i}{p + i} \times 100$$

Now, since both p and i are positive, $\frac{i}{p}$ is greater than $\frac{i}{p + i}$ since the numerator is the same in both fractions and the denominator is lesser and positive. Hence, Statement (2) alone answers the question. The answer to the question is "Yes. Percentage increase was greater in first decade than in second decade."

The answer is (B).

92. The volume of a rectangular tank is *length* × *width* × *depth* = 6 feet × 5 feet × *depth*. Hence, the volume of the conical box, which is 3 times the volume of rectangular box, is $3(6 \times 5 \times depth)$. Since the time taken to fill a tank equals the (volume of the tank) ÷ (the rate of filling), the conical box should be filled in $3(6 \times 5 \times depth)/6$ cubic feet per minute. The only unknown in the expression is the *depth* (of the rectangular box). Hence, Statement (1) alone is sufficient to answer the question. Since the radius of the conical box alone does not determine the volume of the conical box, Statement (2) alone is not sufficient. Hence, the answer is (A).

93. Let each expression in Statement (1) equal k. Then we have the equation $\frac{x}{b-c} = \frac{y}{c-a} = \frac{z}{a-b} = k$. This reduces to $x = (b - c)k$, $y = (c - a)k$, $z = (a - b)k$. Now, $ax + by + cz$ equals

$$a(b - c)k + b(c - a)k + c(a - b)k =$$
$$k(ab - ac + bc - ba + ca - cb) =$$
$$k \times 0 =$$
$$0$$

Hence, Statement (1) alone determines the value of the given expression.

Now, from Statement (2) alone, we have $ax : by : cz = 1 : 2 : -3$. Let $ax = t$, $by = 2t$, and $cz = -3t$ (such that $ax : by : cz = 1 : 2 : -3$). Then $ax + by + cz = t + 2t - 3t = 0$. Hence, Statement (2) alone is also sufficient.

Since both statements are individually sufficient, the answer is (D).

94. From Statement (1) alone, we have that the ratio of the three angles of the triangle is 1 : 2 : 3. Let $k°$, $2k°$, and $3k°$ be the three angles. Summing the three angles to 180° yields $k + 2k + 3k = 180$; $6k = 180$; $k = 180/6 = 30$. Now that we have the value of k, we can calculate the three angles and determine whether the triangle is a right triangle or not. Hence, Statement (1) alone is sufficient. The third angle equals $3k = 3 \times 30° = 90°$ and therefore the triangle is right angled. Hence, the answer to the question is "Yes. The triangle is right angled."

From Statement (2) alone, we have that one angle of the triangle equals the sum of the other two angles. Let $\angle A = \angle B + \angle C$. Summing the angles of the triangle to 180° yields $\angle A + \angle B + \angle C = 180$; $\angle A + \angle A = 180$ (since $\angle B + \angle C = \angle A$); $2\angle A = 180$; $\angle A = 180/2 = 90$. Hence, the triangle is a right triangle, and therefore Statement (2) alone is also sufficient.

Hence, the answer is (D).

95. We are given that the radius of the circle is 3 units and BC, which is tangent to the circle, is a side of the square $ABCD$. Now, let's test if we will be able to construct a figure with unique dimensions for the problem, using the statements.

From Statement (1) alone, we have that $PC = 3$. From the figure, OC equals

$$[OP \text{ (radius of circle)}] + [PC \text{ (= 3 units, given)}] =$$
$$\text{radius} + 3 =$$
$$3 + 3 =$$
$$6$$

Now, draw a circle with radius OB of length 3 units (given that the radius is 3 units). Then draw a perpendicular (tangent) to the line from the point B. Then locate a point on this line that is 6 units from the center O of the circle. The point thus formed is C, and the side length of the square is BC. We can measure this and calculate the area as BC^2. The result is the answer of the question. Hence, Statement (1) alone is sufficient to answer the question.

From Statement (2) alone, we have that $x = 60$. Hence, draw a circle with radius OB and radius 3. Then draw a perpendicular to OB from the point B. Then draw a line making 60° to meet the tangent. Name the point of intersection C. Now, BC is the side length of the square $ABCD$ and BC^2, the area. Hence, calculate BC and then evaluate the area. The result is the answer. Hence, Statement (2) alone is also sufficient to answer the question.

Hence, the answer is (D).

96. Name the first triangle ABC and the second triangle DEF. From statements (1) and (2) together, let $\angle A : \angle B : \angle C = 2 : 3 : 4$, and let $\angle D : \angle E : \angle F = 4 : 3 : 2$. The ordering of the ratio can be reversed and doing this for the second ratio equation yields $\angle F : \angle E : \angle D = 2 : 3 : 4$. Now since the ratios of the corresponding angles in triangle ABC equals the ratio of angles in triangle DEF, we have that triangles ABC and FED are similar. The statements can together answer the question. Hence, the answer is (C).

97. The angle ratio of the triangle given in Statement (1) is 2 : 3 : 4 which in any order is not equal to the ratio of angles of the triangle in Statement (2), which is 1 : 1 : 1. Hence, the statements together say that the two triangles are not similar. We need both statements (1) and (2) to solve it. Hence, the answer is (C).

Method II: The ratio of the angles of the triangle in Statement (1) is 2 : 3 : 4. So, let the three angles of the triangle be $2x$, $3x$, and $4x$. Since the sum of the angles of a triangle is 180°, $2x + 3x + 4x = 180$. Solving this equation for x yields $x = 180/9 = 20$. Hence, the three angles are $2x = 2 \times 20 = 40$, $3x = 3 \times 20 = 60$, and $4x = 4 \times 20 = 80$.

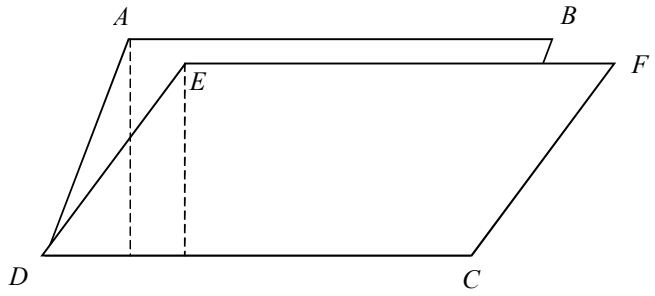
Now, the ratio of the angles of the triangle in Statement (2) is 1 : 1 : 1. Let the three angles of the triangle be y, y, and y. Summing the three to 180° yields $y + y + y = 180$, so $3y = 180/3 = 60$. Hence, each of the three angles in the triangle is 60°.

Thus, not all three angles of the two triangles are equal. Hence, the two triangles are not similar. We need both statements (1) and (2) to solve it. Hence, the answer is (C).

98. From Statement (1), we have that sides AB and CD (the two sides are opposite in the quadrilateral $ABCD$ since the two do not have a common vertex) are both equal to 3. Also, from Statement (2), we have that sides AD and BC (the two sides are opposite in the quadrilateral $ABCD$ since the two do not have any a common vertex) are both equal to 4.

Clearly, the statements individually cannot determine the area. So, the answer is not (A), or (B), or (D).

Such a figure, in which pairs of opposite sides are equal, must be a parallelogram, as shown in the figure below. To evaluate the area of the parallelogram, we also need the inclination of the parallelogram with respect to its base. For example, in the figure shown below, even if AD equals ED, AB equals EF, and BC equals FC, the area of the parallelogram $ABCD$ is greater than area of the parallelogram $EFCD$ since the inclination of $ABCD$ is lesser than that of $EFCD$. The answer is (E).



99. x is not equal to 0 because the statements are not satisfied with the value of 0.

Dividing the inequality in Statement (1) by x^2 yields $x^4 > 1$. Now, whether x is positive or negative, x^4 is positive. Hence, we cannot determine whether x is positive or negative from this inequality alone. So, we have a double case with Statement (1) alone and therefore Statement (1) alone is not sufficient to answer the question: Is x greater than 0.

Now, dividing the inequality in Statement (2) by the positive value x^2 yields $x^3 < 1$. Hence, the range of x is, x is less than 1 (not including 0). There are numbers greater than 0 and less than 0 in the range. So, we have a double case and Statement (2) alone is not sufficient.

From Statement (1) alone, we have $x^4 > 1$. Hence, x must be either greater than 1 or less than –1. From Statement (2), we have $x^3 < 1$. Hence, x must be less than 1 not including 0. The common range that can be derived from the two statements is $x < -1$. Any such number is negative. Hence, the statements together answer the question. The answer is (C).

100. The formula for the volume of a rectangular tank is (*area of base*) × *height*. Hence, we have

volume = (*area of base*) × *height*
or
height = *volume*/(*area of base*)

This equation shows that we need the area of the base and the volume of the tank to calculate its height. The area of the base is given in Statement (1), but Statement (1) alone does not yield the volume of the tank. So, we just need the volume of the tank to calculate the height (to answer the question). Now, from Statement (2), we are given that it takes 20 seconds to fill the tank at the rate of 25 cubic feet per second. Hence, the volume of the tank = rate × time = 25 cubic feet × 20 seconds = 500 cubic feet. Using this, the height can be calculated and therefore the question can be answered. Since statements (1) and (2) together are sufficient to answer the question, the answer is (C).

101. Statements (1) and (2) individually are not sufficient to determine the value of $x^3 + y^3$ because in each case there are two variables but only one equation.

Together the statements form the following system:

$$x + y = 7$$
$$x^2 + y^2 = 25$$

Solving the top equation for y yields $y = 7 - x$. Substituting this into the bottom equation yields

$$x^2 + (7 - x)^2 = 25$$
$$x^2 + 49 - 14x + x^2 = 25$$
$$2x^2 - 14x + 24 = 0$$
$$x^2 - 7x + 12 = 0$$
$$(x - 3)(x - 4) = 0$$
$$x - 3 = 0 \text{ or } x - 4 = 0$$
$$x = 3 \text{ or } x = 4$$

Now, if $x = 3$, then $y = 7 - 3 = 4$; and if $x = 4$, then $y = 7 - 4 = 3$. In either case, $x^3 + y^3$ equals $3^3 + 4^3$ ($= 4^3 + 3^3$). Hence, statements (1) and (2) together are sufficient to answer the question. The answer is (C).

102. x is one of the numbers in the set 1/54, 1/23, and 1/12.

From Statement (1) alone, we have the inequality $1/55 < x < 1/22$. Since the numbers 23 and 54 lie between 22 and 55, the reciprocals 1/23 and 1/54 lie between 1/55 to 1/22. Hence, x could be either 1/23 or 1/54. Hence, Statement (1) alone is not sufficient.

From Statement (2) alone, we have the inequality $1/33 < x < 1/11$. Since the numbers 12 and 23 lie between the numbers 11 and 33, the reciprocals 1/12 and 1/23 lie in the range 1/33 to 1/11. Hence, x could be either 1/11 or 1/23. Hence, Statement (2) alone is not sufficient.

Now, from statements (1) and (2) together, we have $1/55 < x < 1/22$ and $1/33 < x < 1/11$. Combining the two inequalities yields $1/33 < x < 1/22$. Since among the three numbers 12, 23, and 54, the number 23 is the only one between 22 and 33, only the number 1/23 (among the three numbers 1/54, 1/23, and 1/12) lies between 1/23 and 1/54. Hence, $x = 1/23$. Since the statements together determine the value of x, the answer is (C).

103. From Statement (2) alone, we have $x = y + 3$. Hence, x is 3 units greater than y. Hence $x > y$. This is the same inequality as in Statement (1). Hence, Statement (1) is a derivation of Statement (2). Hence, if the question can be answered by Statement (1) alone, it should be answered by using Statement (2) alone also.

From Statement (1) alone, we have that $x > y$. Now, we are given that in ΔPQR, $PQ = x$, $QR = x + 3$, and $PR = y$. Since $PQ = x$, $PR = y$, and $x > y$, $PQ > PR$. Also, since $QR = x + 3$, and $PQ = x$, QR equals $PR + 3$. Hence, QR is greater than PQ; $QR > PQ$. Combining the two inequalities $PQ > PR$ and $QR > PQ$ yields $QR > PQ > PR$. So, the longest side the triangle is QR and therefore the greatest angle in it should be the angle opposite the side, the angle P. So, a solution is possible with Statement (1) alone. Hence, the solution is possible with Statement (2) alone as well.

The answer is (D).

104. Let t be the entire time of the trip.

Then from Statement (1) alone, we have that the car traveled at 75 mph for $t/2$ hours and at 40 mph for the remaining $t/2$ hours. Remember that *Distance* = *Speed* × *Time*. Hence, the net distance traveled during the two periods equals $75 \times t/2 + 40 \times t/2$. Now, remember that

$$\textit{Average Speed} =$$

$$\textit{Net Distance} / \textit{Time Taken} =$$

$$\frac{75 \times t/2 \square 40 \times t/2}{t} \square$$

$$75 \times 1/2 + 40 \times 1/2$$

Hence, Statement (1) alone is sufficient.

Now, from Statement (2) alone, we have that if the car had constantly traveled at 75 mph, it would have needed 5 hrs to complete the trip. But, we do not know how much time it has actually taken to complete the trip. Hence, we cannot find the average speed. So, Statement (2) alone is not sufficient.

The answer is (A).

105. From Statement (1) alone, we have that $s/t = 39.13$. But, 39.13 is the result of one of the ratios: $\frac{3913}{100}$, $\frac{2 \times 3913}{2 \times 100}$, $\frac{3 \times 3913}{3 \times 100}$, …, and so on.

The first fraction $\frac{3913}{100}$ equals $\frac{3900}{100} \square \frac{13}{100}$, so 13 is the remainder.

The second fraction $\frac{7826}{200}$ equals $\frac{7800}{200} \square \frac{26}{200}$, so the remainder is 26.

Hence, we have do not have a unique value for the remainder. Hence, Statement (1) alone is not sufficient.

From Statement (2) alone, we have that $s = 17{,}731$ and $t = 429$. Since we have the values of the dividend and the divisor, we can find the quotient and the remainder. Hence, Statement (2) alone is sufficient. Calculating the remainder is not necessary.

The answer is (B).

106. From Statement (1) alone, we have that the cost of production for 300 units is \$300. Suppose the cost of production (in dollars) is calculated as 100 + 2/3 × quantity. Then the cost of production of 300 units equals 300 dollars, and the cost of production of 270 units equals $100 + 2/3 \times 270 = 100 + 180 = 280$.

Now, suppose it were calculated as 200 + 1/3 × quantity, then the cost of production for 270 units would equal $200 + 1/3 \times 270 = 200 + 90 = 290$ dollars.

Since we do not have a unique value, Statement (1) alone is not sufficient.

From Statement (2) alone, the cost of production is proportional to the number of units produced. Hence, we have the equation *The Cost of Production* = $k \times$ *Quantity*, where k is some constant. The *Cost of Production* of 270 units equals $k \times 270$. But we do not know the value of k. So, Statement (2) alone is not sufficient.

Now, from the statements together, we have that 300 units cost 300 dollars. Putting this in the proportionality equation yields $300 = k \times 300$. Solving the equation for k yields $k = 300/300 = 1$. Since we now know the value of k, we can evaluate the cost of production of 270 items. Hence, the statements together answer the question. Hence, the answer is (C). The answer is the cost = $k \times 270 = 1 \times 270 = 270$ dollars.

107. From Statement (1) alone, we have that $x = y = z = p$. Each angle in a quadrilateral is equal when the quadrilateral is a rectangle. A rectangle (only opposite sides necessarily equal) may or may not be a square (all sides equal). Hence, Statement (1) alone is not sufficient to answer the question.

From Statement (2) alone, we have that $AB = BC = CD = DA$. All the sides in a quadrilateral are equal when the quadrilateral is a rhombus. Since a rhombus (only opposite angles necessarily equal) may or may not be a square (all angles equal), Statement (2) alone is not sufficient to answer the question.

Now, from the statements together, we have that in the quadrilateral all the angles are equal and all the sides are equal. This is possible only in a square. Hence, both statements are required to answer the question. The answer is (C).

108. In the figure, points $A(0, 2)$ and $D(0, 0)$ have the same x-coordinate (which is 0). Hence, the two points must be on the same vertical line in the coordinate system.

Similarly, the x-coordinates of the points B and C are the same (both equal a). Hence, the points are on the same vertical line in the coordinate system.

Since any two vertical lines are parallel, lines AD and BC are parallel (for any value of a).

Now, if $ABCD$ is a parallelogram, then the opposite sides must also be equal. Hence, AD must equal BC.

Since AD and BC are vertical lines, AD equals the y-coordinate difference of points A and $D = 2 - 0 = 2$, and BC equals the y-coordinate difference of points B and $C = b - 2$. Now, AD equals BC when $b - 2 = 2$, or $b = 4$. This is when $ABCD$ is a parallelogram and in any other case (when $b \neq 4$), it is not a parallelogram. So, we need the value of b [value given in Statement (2)] alone and the value of a is unimportant here. So, Statement (1) is not needed. Since $b = 5$ in Statement (2) (which shows that $b \neq 4$ and therefore $ABCD$ is not parallelogram), the statement is sufficient. Hence, the answer is (B).

109. In the figure, x equals y (vertical angles are equal); and if AB is perpendicular to CD, then the angles x and y must each equal 90°.

Now, from Statement (1) alone, we have that $x + y = 180$; $x + x = 180$ (since $x = y$); $2x = 180$; $x = 180/2 = 90$. Since we have the value of x, from Statement (1) alone the statement is sufficient. The answer to the question is "Yes. AB is perpendicular to CD."

From Statement (2) alone, we have that AOC is an isosceles triangle. Since x equals y (earlier result) and neither equals 0°, $\angle A$ (= x°) cannot equal $\angle AOC$ (which actually equals $y/2$). Hence, A and O are not the equal angles of the triangle.

Now, suppose $\angle C = \angle AOC$ (hence, isosceles). Then $\angle C = \angle AOC = x$°, and summing the angles of the triangle to 180° yields $\angle AOC + \angle C + \angle A = 180$; $x + x + y/2 = 180$ (by substituting known values); $x + x + x/2 = 180$ (since $y = x$). Solving the equation for x yields $x = 72$. Here, x is not 90°, hence, AB is not perpendicular to CD.

Now, suppose $\angle C = \angle A$ (hence, the triangle is isosceles). Then $\angle C = \angle A = y/2$. Summing the angles of the triangle AOC to 180° yields $\angle AOC + \angle C + \angle A = 180$; $x + y/2 + y/2 = 180$; $x + y = 180$; $x + x = 180$ (since $y = x$); $2x = 180$; $x = 180/2 = 90$. Here, x equals 90°, hence, AB is perpendicular to CD.

Hence, we have a double case, and Statement (2) alone is not sufficient.

Hence, the answer is (A).

110. A unique regular hexagon can be drawn given the radius of the circle in which it is inscribed. The area can be later measured. Hence, Statement (1) alone is sufficient.

The perimeter of the hexagon is $40\sqrt{3}$. Hence, the sum of the 6 equal sides is $40\sqrt{3}$, and the side length of each is $\frac{40\sqrt{3}}{6}$. Given the side length of a regular hexagon, a unique figure can be drawn for the hexagon. The area of the figure can then be measured. Hence, Statement (2) alone is also sufficient.

Hence, the answer is (D).

111. From Statement (1) alone, we have that ACD is a right triangle. So, one of the three angles of the triangle $\angle CAD$, $\angle ACD$, or $\angle D$ is a right angle. If $\angle D$ were the right angle, then since $\angle BCA = \angle D + \angle DAC$ (exterior angle of a triangle equals the sum of the interior opposite angles), $\angle BCA = 90° + \angle DAC$, clearly greater than 90°. Hence, $\angle BCA$ would be greater than 90° (obtuse angle). Hence, it's not a right triangle. But in case $\angle ACD$ is the right angle, then since the angle in a line is 180°, we have $\angle ACD + \angle BCA = 180°$; $90° + \angle BCA = 180°$; $\angle BCA = 90°$. Here, triangle ABC is a right triangle. Hence, we have a double case, and Statement (1) alone is not sufficient.

From Statement (2) alone, we have that AC is the greatest side of ΔACD. This only confirms that the angle opposite side $\angle D$ is greater than 60° and therefore $\angle BCA$ is still open to be a right angle or not.

Now, from the statements together, AC is the greatest side of ΔACD (Hence, $\angle D$ is the greatest angle of the triangle) and the triangle is a right triangle. Since in a right triangle, the greatest angle is the right angle, $\angle D$ is the right angle. Now, $\angle BCA = \angle D + \angle DAC$ (The exterior angle of a triangle equals the sum of the interior opposite angles); $\angle BCA = 90° + \angle DAC$, clearly greater than 90°. Hence, $\angle BCA$ is greater than 90°, and triangle ABC is an obtuse triangle. Hence, triangle ABC is not a right triangle. Since we needed both statements (1) and (2) to answer the question, the answer is (C).

112. Subtracting $(x - y)^2$ from both sides of $(x - y)^3 > (x - y)^2$ in Statement (1) yields

$$(x - y)^3 - (x - y)^2 > 0$$

Factoring out $(x - y)^2$ yields

$$(x - y)^2(x - y - 1) > 0$$

Since is $(x - y)^2$ positive (being a square), $x - y - 1$ must be positive, otherwise $(x - y)^2(x - y - 1)$ would be negative. Hence, $x - y - 1 > 0$, or $x > y + 1$. This inequality says that x is at least one unit larger than y, so $x > y$. Hence, Statement (1) alone is sufficient.

Now, dividing both sides of the inequality $(x - y)^4 > (x - y)^3$ in Statement (2) by the positive expression $(x - y)^2$ yields $(x - y)^2 > x - y$. This inequality is true when either $x - y$ is negative or $x - y$ is positive and greater than 1. If $x - y$ is negative, x is less than y; and if $x - y$ is positive, x is greater than y. Hence, we have a double case, and therefore Statement (2) alone is *not* sufficient to answer the question.

The answer is (A).

113. This problem can be solved with a series. Let the payments for the 7 continuous days be a_1, a_2, a_3, …, a_7. Since each day's pay was 11 dollars more than the previous day's pay, the rule for the series is $a_{n+1} = a_n + 11$.

If we are able to find all 7 numbers in the sequence, we can add the numbers and answer the question. In order to find all the numbers, we need at least one element in the series.

By the rule, let the payments for each consecutive day be listed as

a_1
$a_2 = a_1 + 11$
$a_3 = a_2 + 11 = (a_1 + 11) + 11 = a_1 + 22$
$a_4 = a_3 + 11 = (a_1 + 22) + 11 = a_1 + 33$
$a_5 = a_4 + 11 = (a_1 + 33) + 11 = a_1 + 44$
$a_6 = a_5 + 11 = (a_1 + 44) + 11 = a_1 + 55$
$a_7 = a_6 + 11 = (a_1 + 55) + 11 = a_1 + 66$

From Statement (1) alone, we have that the net pay for the first 4 days equals the net pay for the last 3 days.

The net pay for the first 4 days is $a_1 + (a_1 + 11) + (a_1 + 22) + (a_1 + 33) = 4a_1 + 11(1 + 2 + 3)$.

The net pay for last (next) 3 days is $(a_1 + 44) + (a_1 + 55) + (a_1 + 66) = 3a_1 + 11(4 + 5 + 6)$.

Equating the two yields

$$4a_1 + 11(1 + 2 + 3) = 3a_1 + 11(4 + 5 + 6)$$

This equation can be solved for a_1. Then all the numbers in the series can be calculated, and all the numbers summed to evaluate his net pay. Hence, Statement (1) alone is sufficient.

Now, from Statement (2) alone, we have that he was paid 330 dollars more for his last 3 days of work than he was paid for his first 3 days of work. The net pay for first 3 days is $a_1 + (a_1 + 11) + (a_1 + 22) = 3a_1 + 11(1 + 2)$.

The net pay for last 3 days is $(a_1 + 44) + (a_1 + 55) + (a_1 + 66) = 3a_1 + 11(4 + 5 + 6)$.

Hence, the difference between the two is

$$3a_1 + 11(4 + 5 + 6) – [3a_1 + 11(1 + 2)] =$$

$$3a_1 + 11(4 + 5 + 6) – 3a_1 – 11(1 + 2) =$$

$11(4 + 5 + 6) – 11(1 + 2)$, and this equals 66 dollars.

Hence, Statement (2) is an inference from the information given in the problem description and is not new information. So, the statement is not needed.

The answer is (A).

114. From Statement (1) alone, we have that the ratio of the sum of the reciprocals of x and y to the product of the reciprocals of x and y is 1 : 3. Writing this as an equation yields

$$\frac{\frac{1}{x}+\frac{1}{y}}{\frac{1}{x}\cdot\frac{1}{y}}=\frac{1}{3}$$

$$\frac{\frac{x+y}{xy}}{\frac{1}{xy}}=\frac{1}{3}$$

$$\frac{x+y}{xy}\cdot\frac{xy}{1}=\frac{1}{3}$$

$x+y=\frac{1}{3}$ by canceling xy from the numerator and denominator

Hence, Statement (1) alone is sufficient.

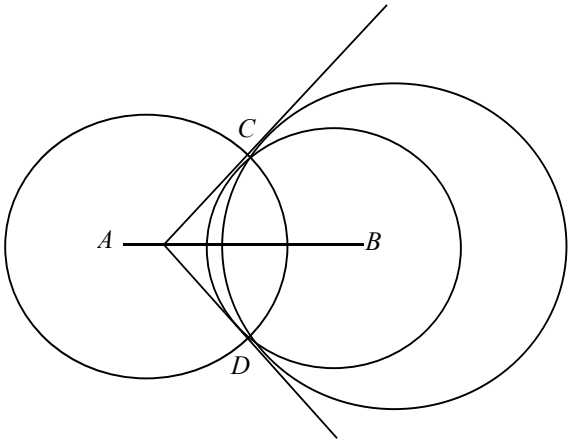
From Statement (2) alone, we have that the product of x and y is 11/36 units greater than the sum of x and y. Writing this as an equation yields

$xy-(x+y)=11/36$
$x+y=xy-11/36$ by solving for $x+y$

Since we do not know the value of xy, we cannot determine the value of $x+y$. Hence, Statement (2) alone is not sufficient.

The answer is (A).

115. Draw a circle of radius 5. So, Statement (1) would be satisfied. Now, from its center, say A, locate two points C and D which subtend an angle 120° with the center ($\angle CAD = 120°$). Now, any circle passing through the points C and D (any one in the figure shown) satisfies the two statements along with the main circle. Each one has a different common area with the main circle. So, the two statements together do not yield a unique value for the common area and the answer is (E), the two statements together is not sufficient.



116. From Statement (1) alone, we have that the product of the three numbers a, b, and c is a multiple of 27. The number 162 (= 27 · 6) is divisible by 81, while the number 54 (= 27 · 2) is not divisible by 81. Since we have a double case, Statement (1) alone is not sufficient.

From Statement (2) alone, we have that none of the three numbers a, b, and c is divisible by 9. Now, even assuming that the three numbers a, b, and c are divisible by at least 3, though not 9, the three numbers can be represented as $3p$, $3q$, and $3r$, where none of the three supposed positive integers p, q, and r is divisible by 3. Then the product of the three numbers equals $3p \cdot 3q \cdot 3r = 27(pqr)$. None of the three numbers p, q, and r is divisible by 3. Hence, pqr is not divisible by 3. So, $27(pqr)$ is not divisible by $27 \cdot 3$, which equals 81. So, the product of abc is not divisible by 81. Hence, Statement (2) alone is sufficient to answer the question.

Hence, the answer is (B), and the answer to the question is "No. abc is not divisible by 81."

117. Let the number of chickens, pigs and horses on Richard's farm be c, p, and h. From Statement (1), we have that the ratio of the three is $c : p : h = 33 : 17 : 21$. So, let c equal $33k$, p equal $17k$, and h equal $21k$, where k is some positive integer (such that $c : p : h = 33 : 17 : 21$). Then the total number of pigs and horses is $17k + 21k = 38k$; and since this is greater than $33k$, the number of chickens, the answer to the question is "The total number of pigs and horses is greater than the number of chickens." Hence, Statement (1) alone is sufficient to answer the question.

Next, Statement (2) alone gives the total number of the three species available, which is of no use. Hence, Statement (2) alone is not sufficient.

Hence, the answer is (A).